

Image Processing (303105381)

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CHAPTER-6

MORPHOLOGY IMAGE PROCESSING

Outline

1. Preliminaries
2. Dilation and Erosion
3. Opening and Closing
4. The Hit-or-Miss Transformation
5. Some Basic Morphological Algorithms.

Outcome of this unit:

After completing the students will be able to:

- To understand the fundamental concepts of Digital Image Processing.
- To understand image compression techniques.
- To understand morphological techniques for storage and transmission purpose.

Introduction

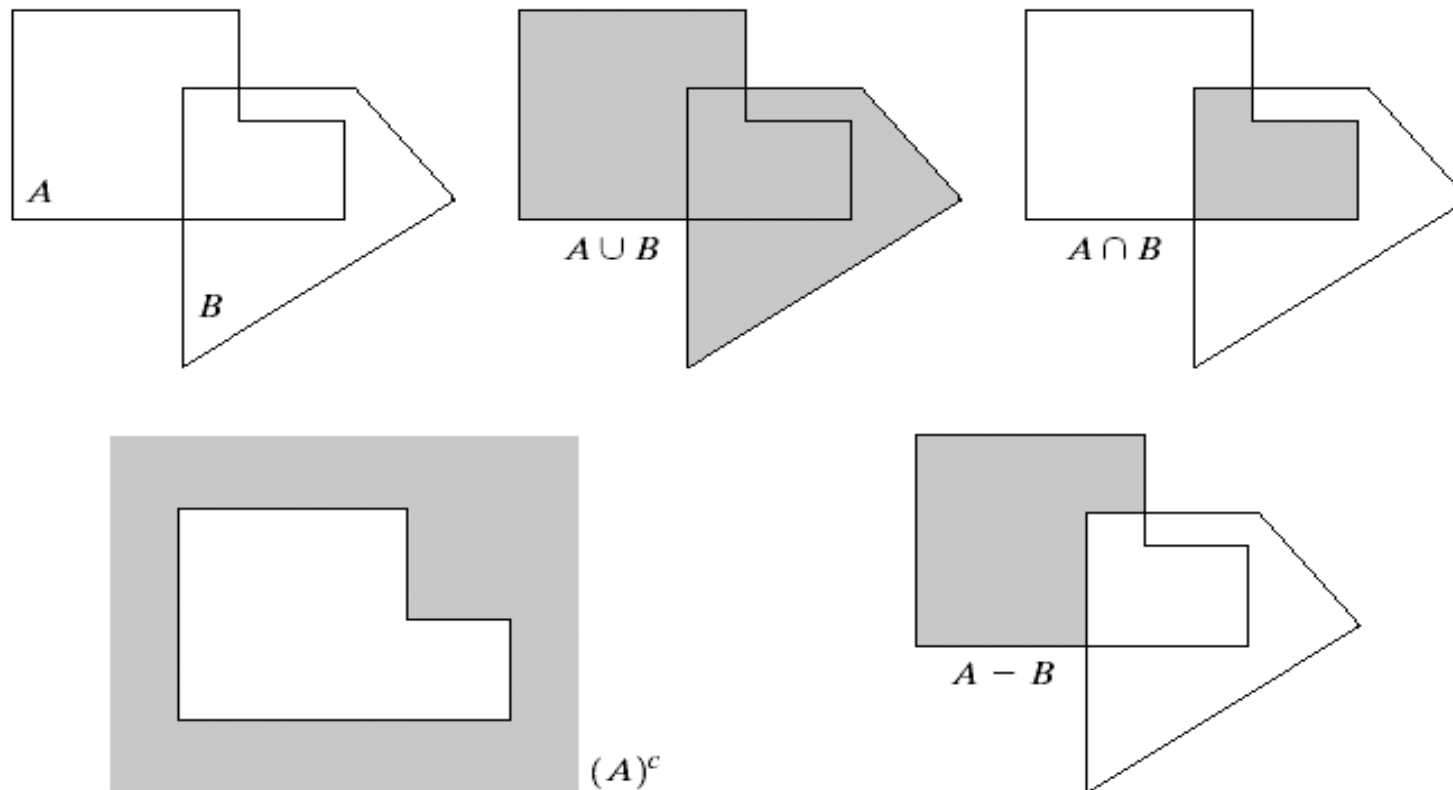
Morphology: A branch of biology that deals with the form and structure of animals and plants

Morphological image processing is used to extract image components for representation and description of region shape, such as boundaries, skeletons, and the convex hull

Introduction to **Morphological Operators**

- ☐ Used generally on binary images, e.g., background subtraction results!
 - ☐ Used on gray value images, if viewed as a stack to binary images.
- Good for, e.g.,
- ☐ Noise removal in background
 - ☐ Removal of holes in foreground / background

Some Basic Concepts from Set Theory

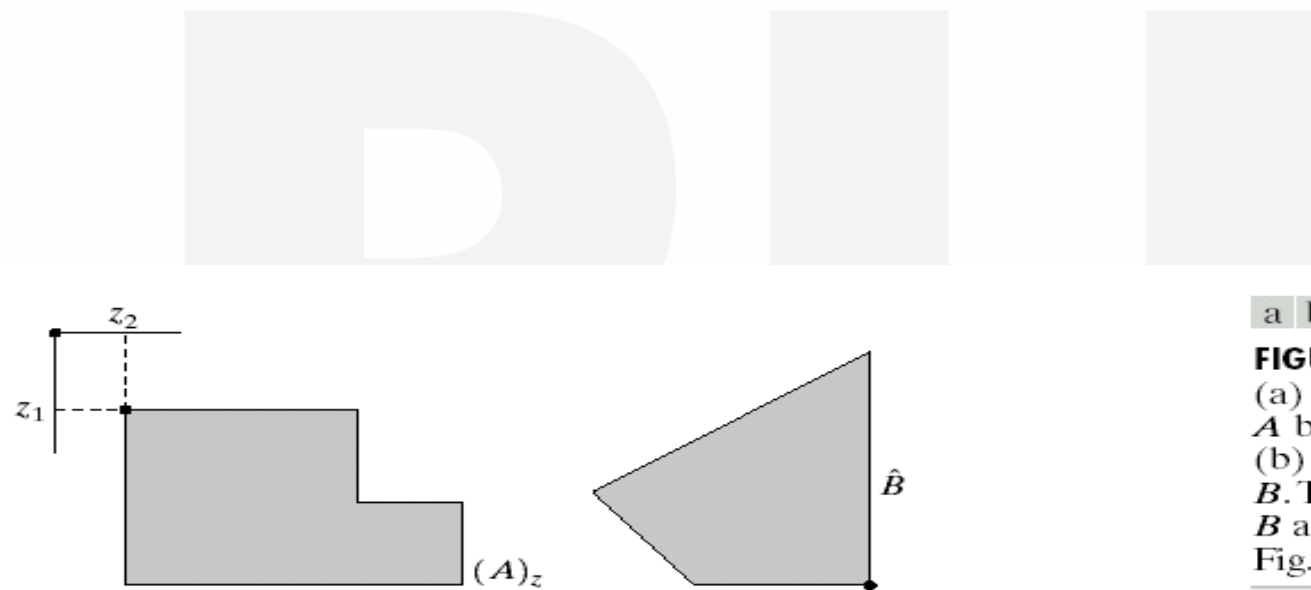


a	b	c
d	e	

FIGURE 9.1
(a) Two sets A and B . (b) The union of A and B . (c) The intersection of A and B . (d) The complement of A . (e) The difference between A and B .



Example



a b

FIGURE 9.2

(a) Translation of A by z .

(b) Reflection of B . The sets A and B are from Fig. 9.1.

Logic Operations Involving Binary Images

TABLE 9.1

The three basic
logical operations.

p	q	$p \text{ AND } q \text{ (also } p \cdot q \text{)}$	$p \text{ OR } q \text{ (also } p + q \text{)}$	$\text{NOT } (p) \text{ (also } \bar{p} \text{)}$
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

Parul

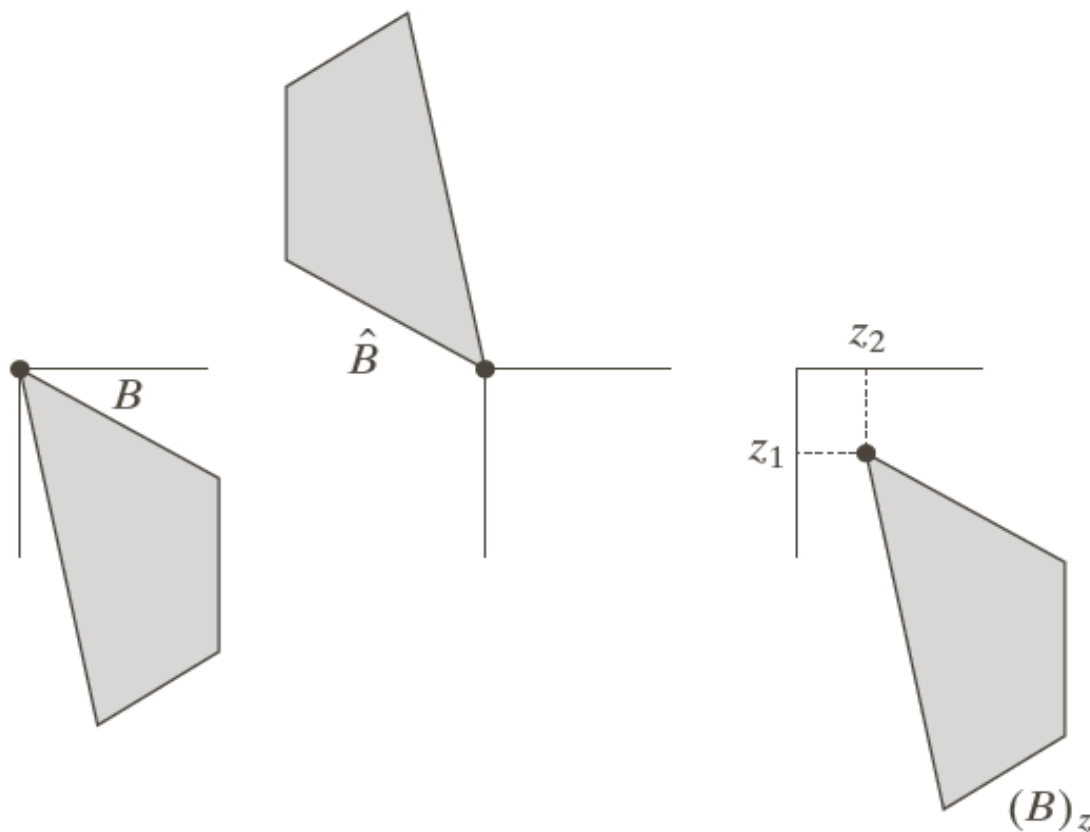


A

B

PU

Example: Reflection and Translation



a b c

FIGURE 9.1

(a) A set, (b) its reflection, and (c) its translation by z .

Preliminaries

Structure elements (SE)

- ☐ Small sets or sub-images used to probe an image under study for properties of interest.
- ☐ Structuring Elements can have varying sizes
- ☐ Usually, element values are 0,1 and none(!)
- ☐ Structural Elements have an origin
- ☐ For thinning, other values are possible
- ☐ Empty spots in the Structuring Elements are don't care's!

Examples: Structuring Elements

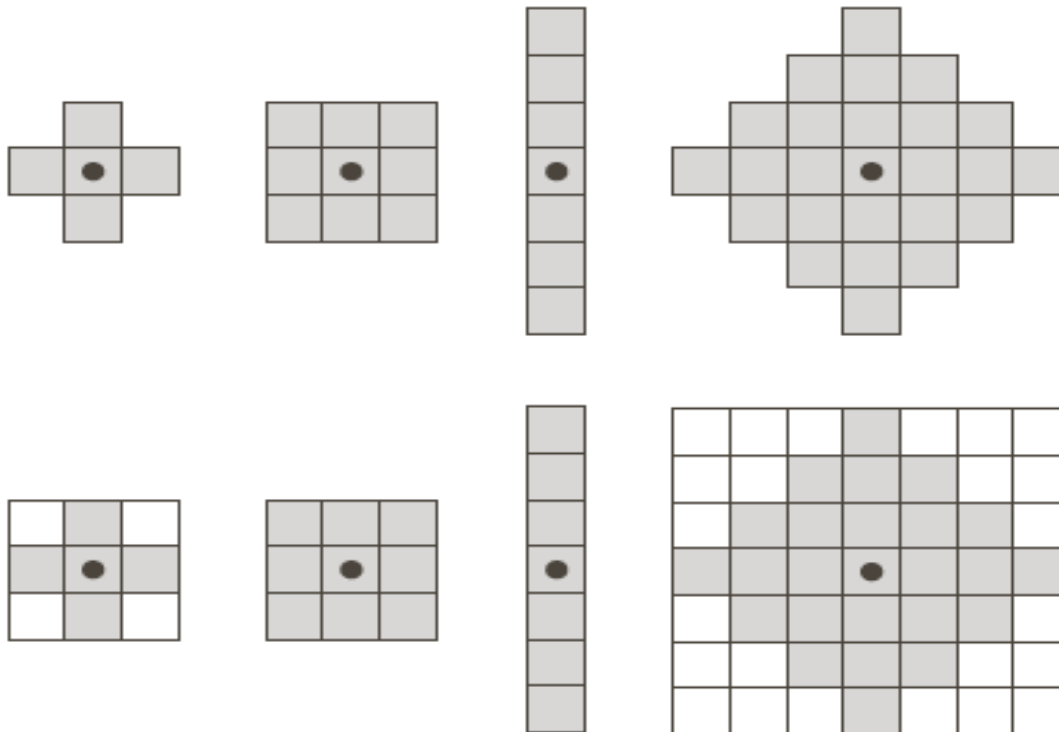
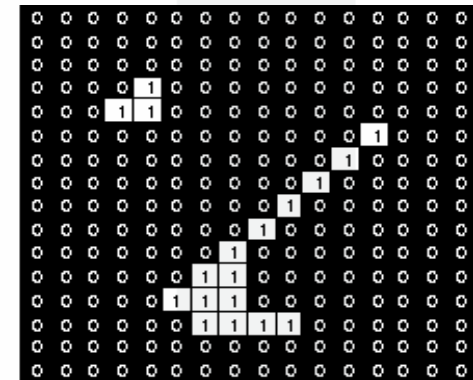
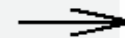
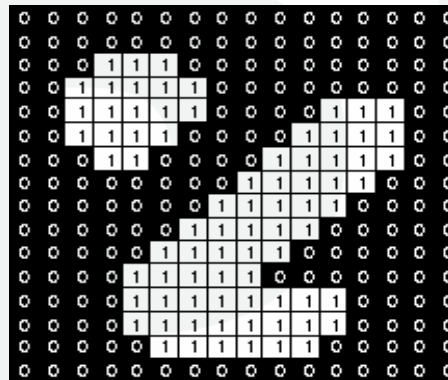


FIGURE 9.2 First row: Examples of structuring elements. Second row: Structuring elements converted to rectangular arrays. The dots denote the centers of the SEs.

A first Example: Erosion

- **Erosion** is an important morphological operation



- Applied **Structuring Element**:

1	1	1
1	1	1
1	1	1

Set of coordinate points =

{ (-1, -1), (0, -1), (1, -1),
(-1, 0), (0, 0), (1, 0),
(-1, 1), (0, 1), (1, 1) }

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0								
--	---	--	--	--	--	--	--	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0							
--	---	---	--	--	--	--	--	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	0						
--	---	---	---	--	--	--	--	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	0	0					
--	---	---	---	---	--	--	--	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	0	0	1				
--	---	---	---	---	---	--	--	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	0	0	1	0			
--	---	---	---	---	---	---	--	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

1	1	1
---	---	---



Output Image

	0	0	0	0	1	0	0		
--	---	---	---	---	---	---	---	--	--

Example for Erosion

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---



Structuring Element

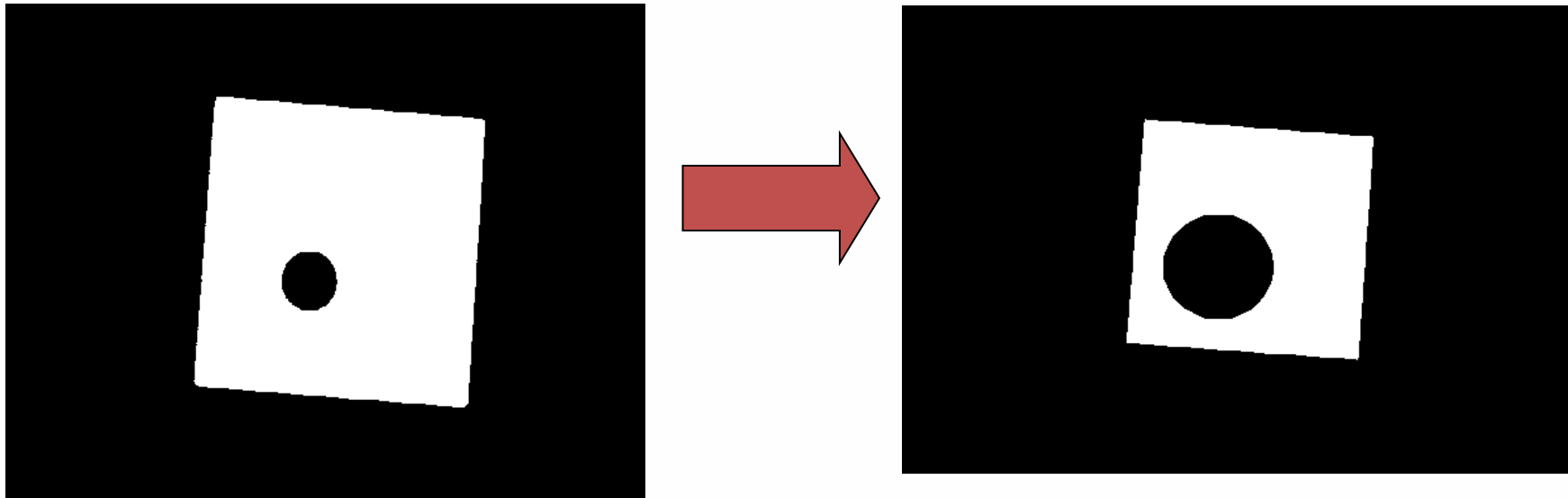
1	1	1
---	---	---



Output Image

	0	0	0	0	1	0	0	0	
--	---	---	---	---	---	---	---	---	--

Another example of erosion

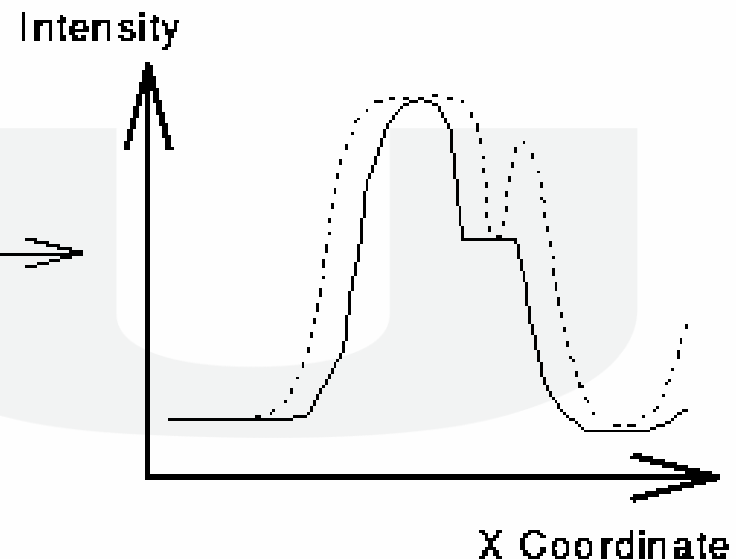
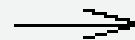
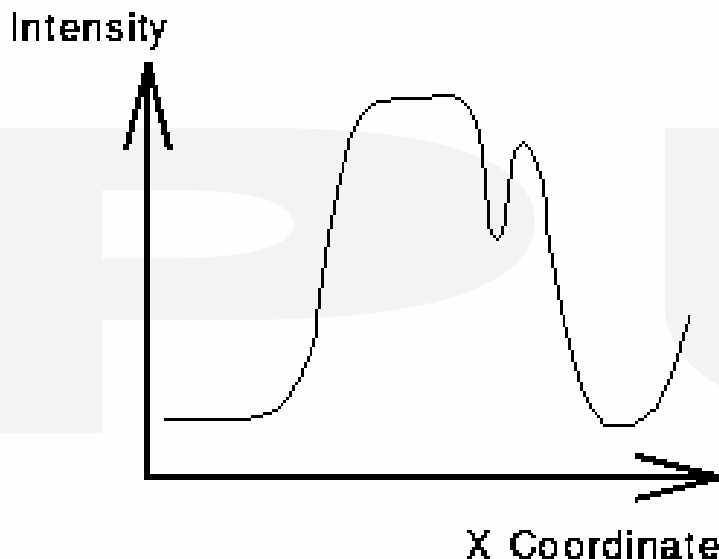


- White = 0, black = 1, dual property, image as a result of erosion gets darker

Erosion on Gray Value Images

View gray value images as a stack of binary images!

Width of eroding kernel: 



Example of Erosion

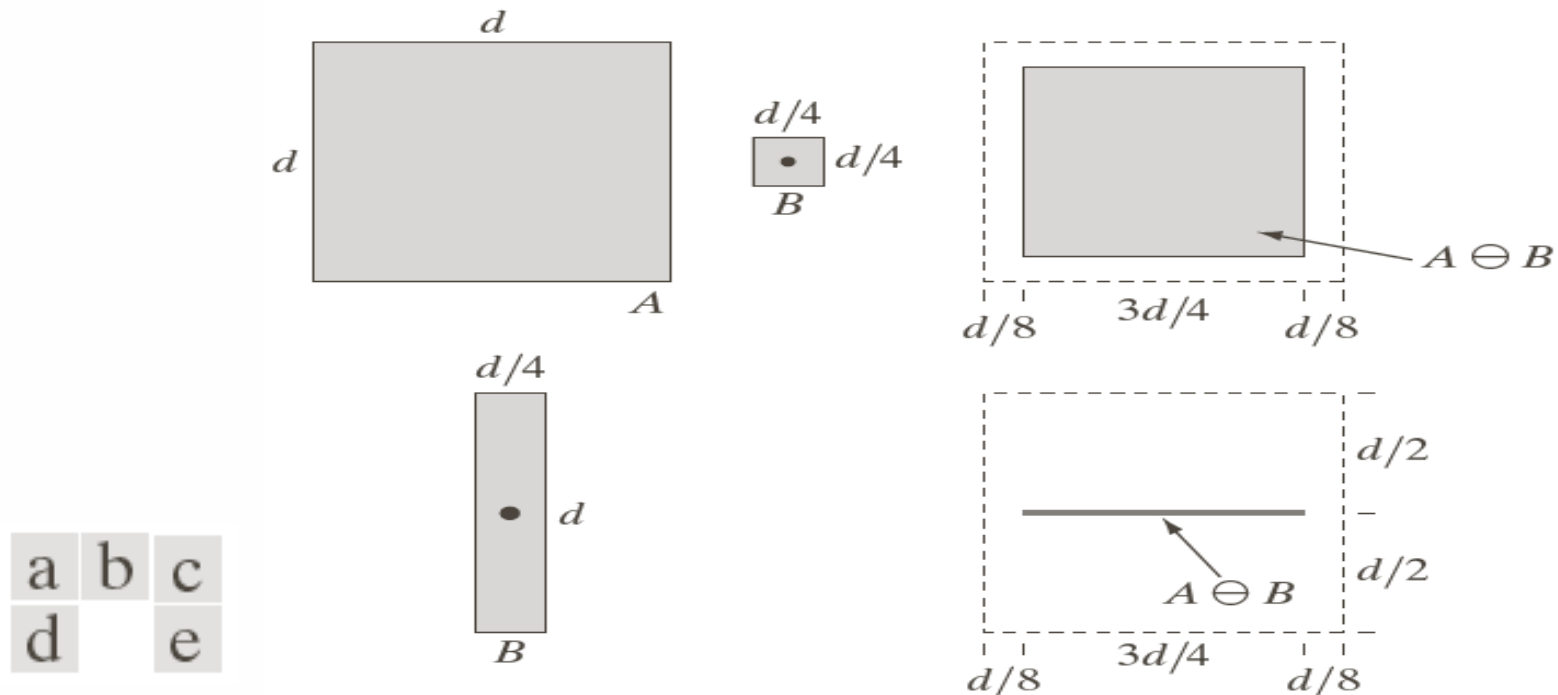
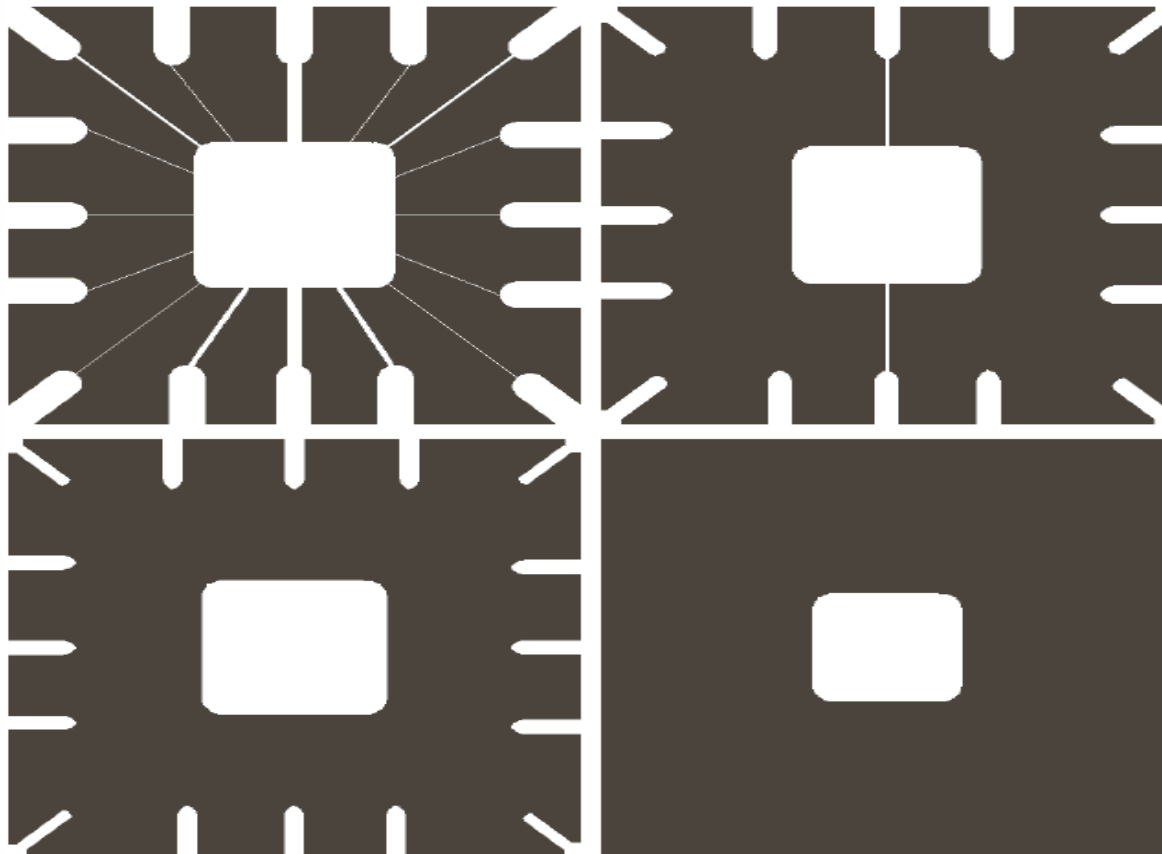


FIGURE 9.4 (a) Set A . (b) Square structuring element, B . (c) Erosion of A by B , shown shaded. (d) Elongated structuring element. (e) Erosion of A by B using this element. The dotted border in (c) and (e) is the boundary of set A , shown only for reference.

Example of Erosion



a	b
c	d

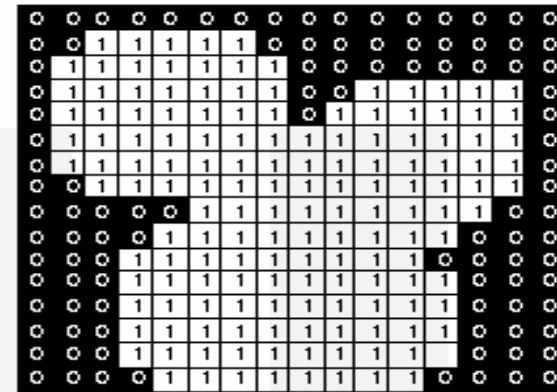
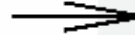
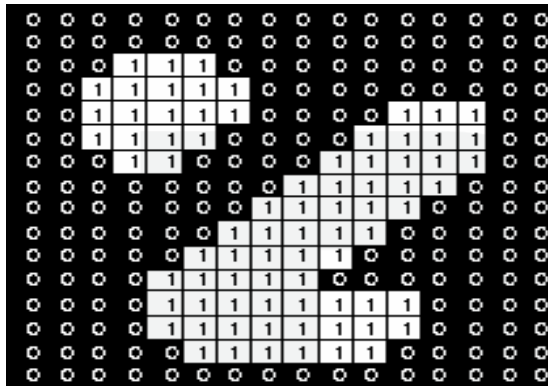
FIGURE 9.5 Using erosion to remove image components. (a) A 486×486 binary image of a wire-bond mask. (b)–(d) Image eroded using square structuring elements of sizes 11×11 , 15×15 , and 45×45 , respectively. The elements of the SEs were all 1s.

Dilation

- ☐ **Dilation** is the set of all points in the image, where the structuring element “touches” the foreground.
- ☐ Consider each pixel in the input image
- ☐ If the structuring element touches the foreground image, write a “1” at the origin of the structuring element!
- ☐ **Input:**
- ☐ **Binary Image**
- ☐ **Structuring Element, containing only 1s!!**

Example: Dilation

Dilation is an important morphological operation



Applied **Structuring Element**:

1	1	1
1	1	1
1	1	1

Set of coordinate points =

{ (-1, -1), (0, -1), (1, -1),
(-1, 0), (0, 0), (1, 0),
(-1, 1), (0, 1), (1, 1) }

Example for Dilation

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---

Structuring Element

1	1	1
---	---	---

Output Image

	1								
--	---	--	--	--	--	--	--	--	--

Example for Dilation

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---

Structuring Element

1	1	1
---	---	---

Output Image

	1	0							
--	---	---	--	--	--	--	--	--	--

Example for Dilation

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---

Structuring Element

1	1	1
---	---	---

Output Image

	1	0	1						
--	---	---	---	--	--	--	--	--	--

Example for Dilation

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---

Structuring Element

1	1	1
---	---	---

Output Image

	1	0	1	1					
--	---	---	---	---	--	--	--	--	--

Example for Dilation

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---

Structuring Element

1	1	1
---	---	---

Output Image

	1	0	1	1	1				
--	---	---	---	---	---	--	--	--	--

Example for Dilation

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---

Structuring Element

1	1	1
---	---	---

Output Image

	1	0	1	1	1	1			
--	---	---	---	---	---	---	--	--	--

Example for Dilation

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---

Structuring Element

1	1	1
---	---	---

Output Image

	1	0	1	1	1	1	1		
--	---	---	---	---	---	---	---	--	--

Example for Dilation

Input image

1	0	0	0	1	1	1	0	1	1
---	---	---	---	---	---	---	---	---	---

Structuring Element

1	1	1
---	---	---

Output Image

	1	0	1	1	1	1	1	1	
--	---	---	---	---	---	---	---	---	--

Another Dilation Example

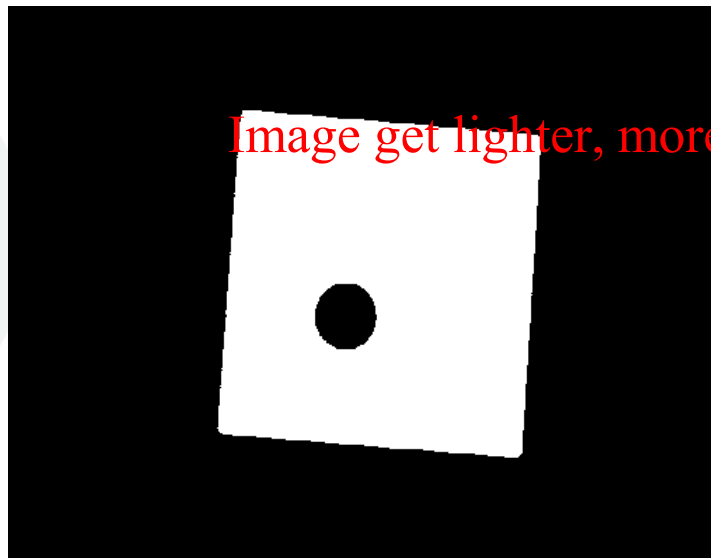
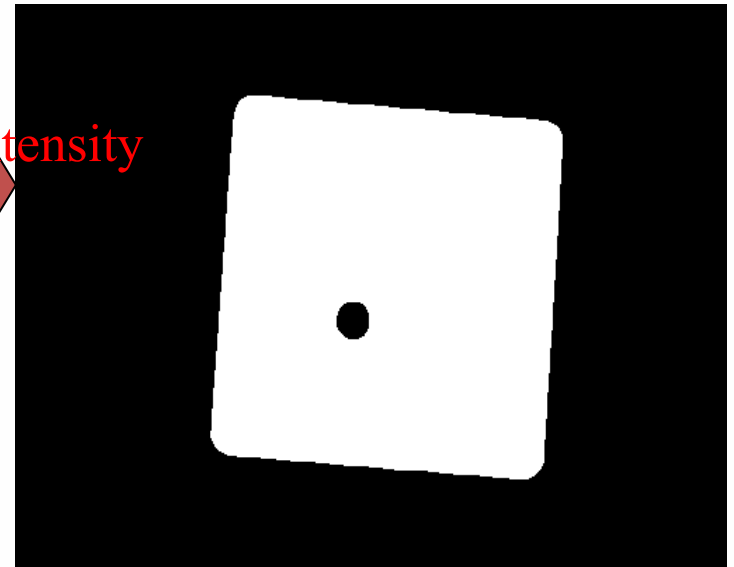
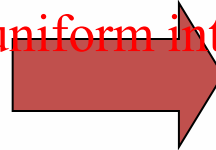
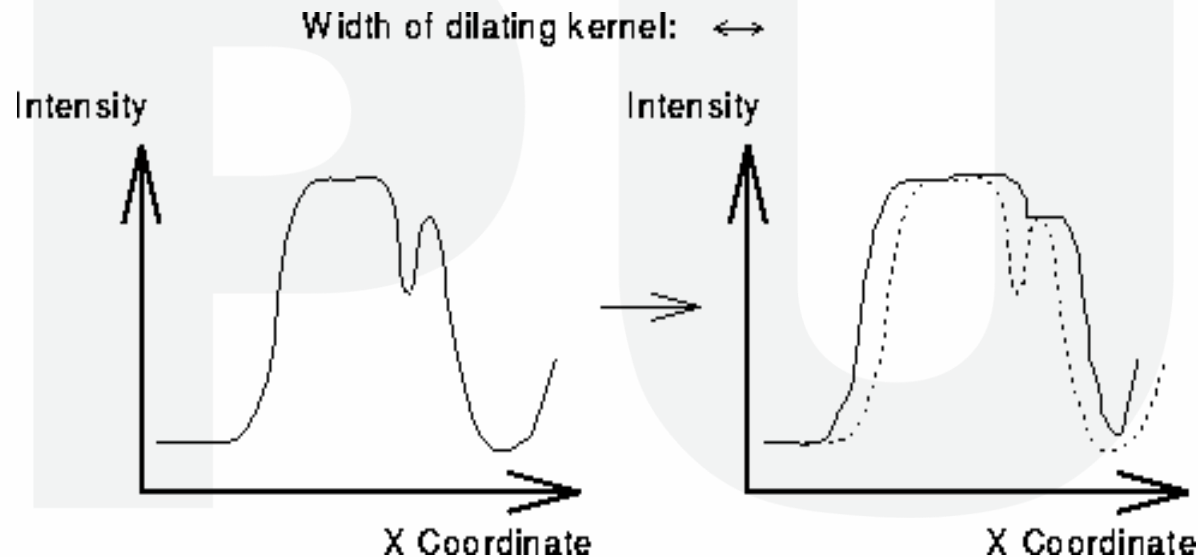


Image get lighter, more uniform intensity

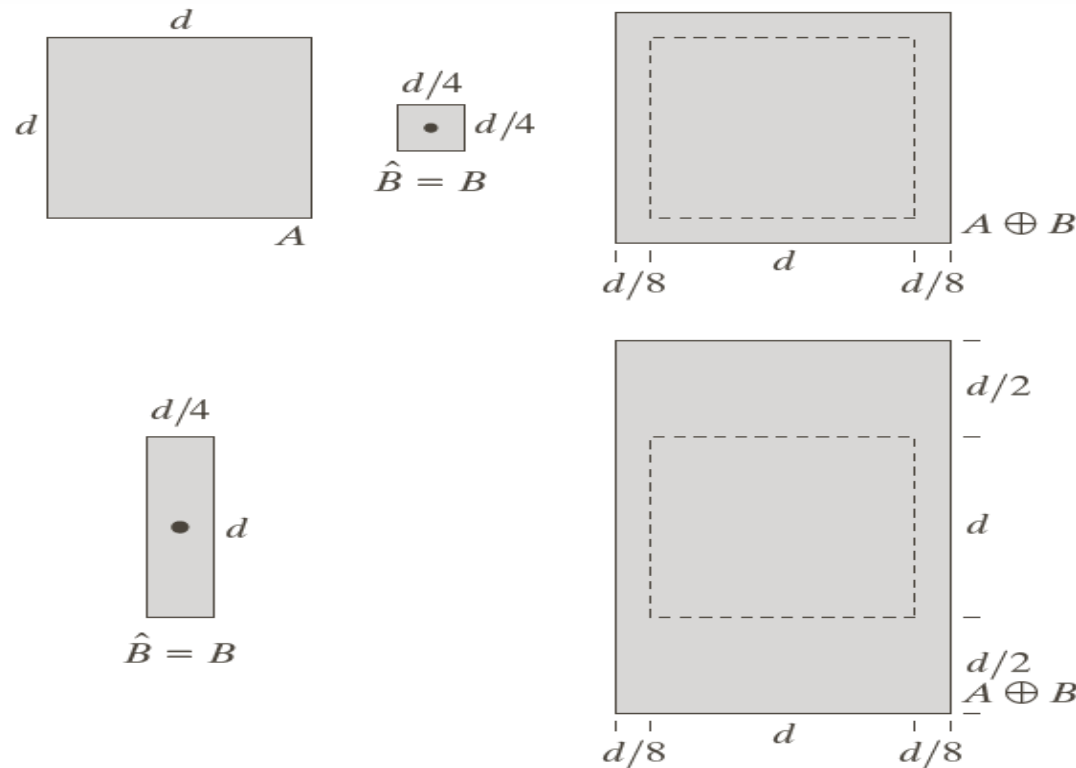


Dilation on Gray Value Images

View gray value images as a stack of binary images!



Examples of Dilation



a	b	c
d		e

FIGURE 9.6

(a) Set A .
 (b) Square structuring element (the dot denotes the origin).
 (c) Dilation of A by B , shown shaded.
 (d) Elongated structuring element.
 (e) Dilation of A using this element. The dotted border in (c) and (e) is the boundary of set A , shown only for reference

Examples of Dilation

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



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0	1	0
1	1	1
0	1	0

a c
b

FIGURE 9.7

(a) Sample text of poor resolution with broken characters (see magnified view).
(b) Structuring element.
(c) Dilation of (a) by (b). Broken segments were joined.

Summary effects of dilation:

- ❑ Expand/enlarge objects in the image
- ❑ Fill gaps or bays of insufficient width
- ❑ Fill small holes of sufficiently small size
- ❑ Connects objects separated by a distance less than the size of the window

Duality

- ☐ Erosion and dilation are duals of each other with respect to set complementation and reflection

Duality

Erosion and dilation are duals of each other with respect to set complementation and reflection

PPU

The erosion of A by B is the set of all points z , such that B , translated by z is contained in A .

3.Opening

Smooth the contour of an object, break narrow isthmuses, and eliminate thin protrusions.

The opening A by B is the erosion of A by B , followed by a dilation of the result by B

Example: Opening

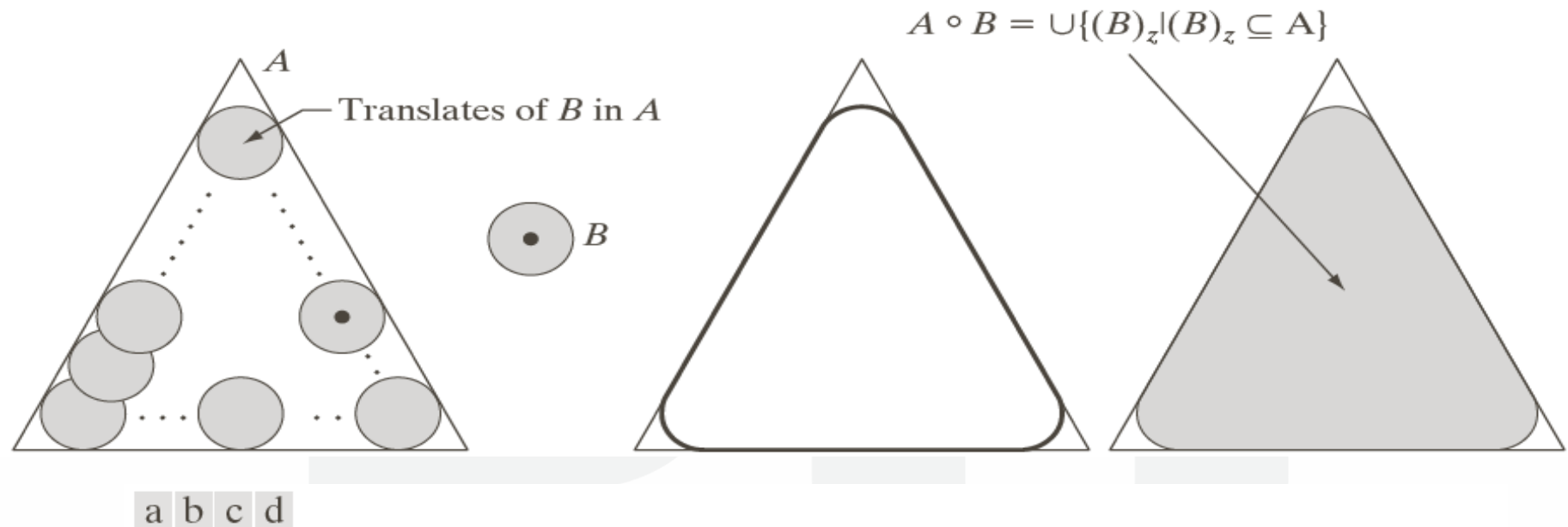


FIGURE 9.8 (a) Structuring element B “rolling” along the inner boundary of A (the dot indicates the origin of B). (b) Structuring element. (c) The heavy line is the outer boundary of the opening. (d) Complete opening (shaded). We did not shade A in (a) for clarity.

Example: Closing

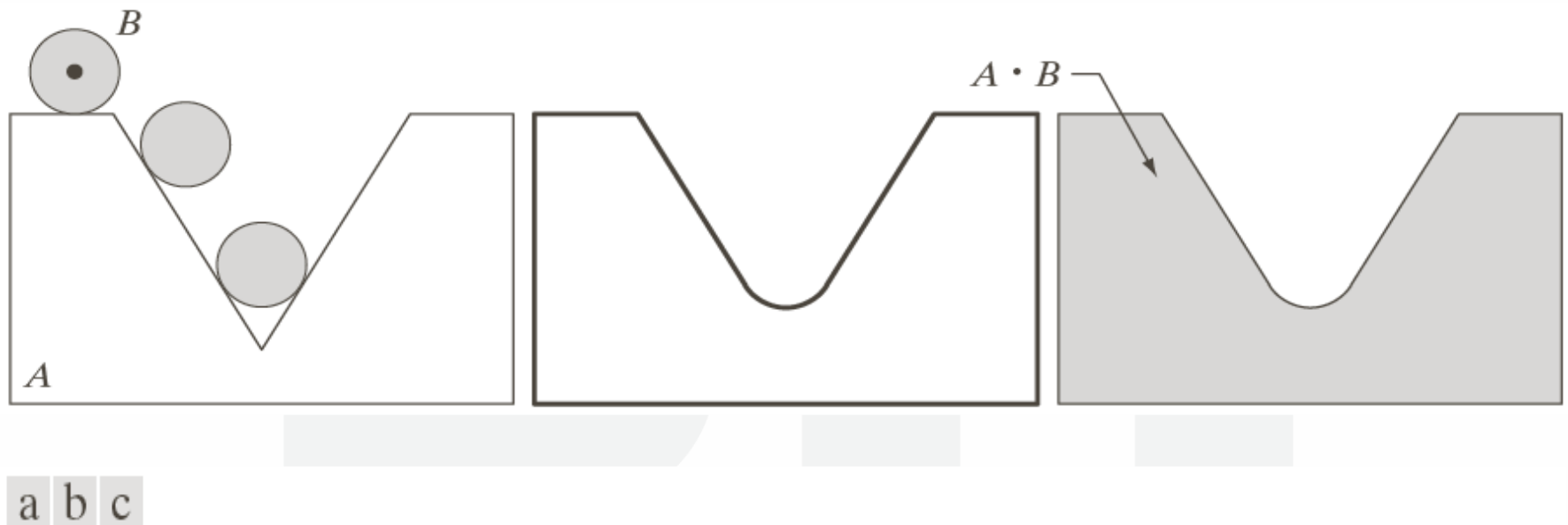
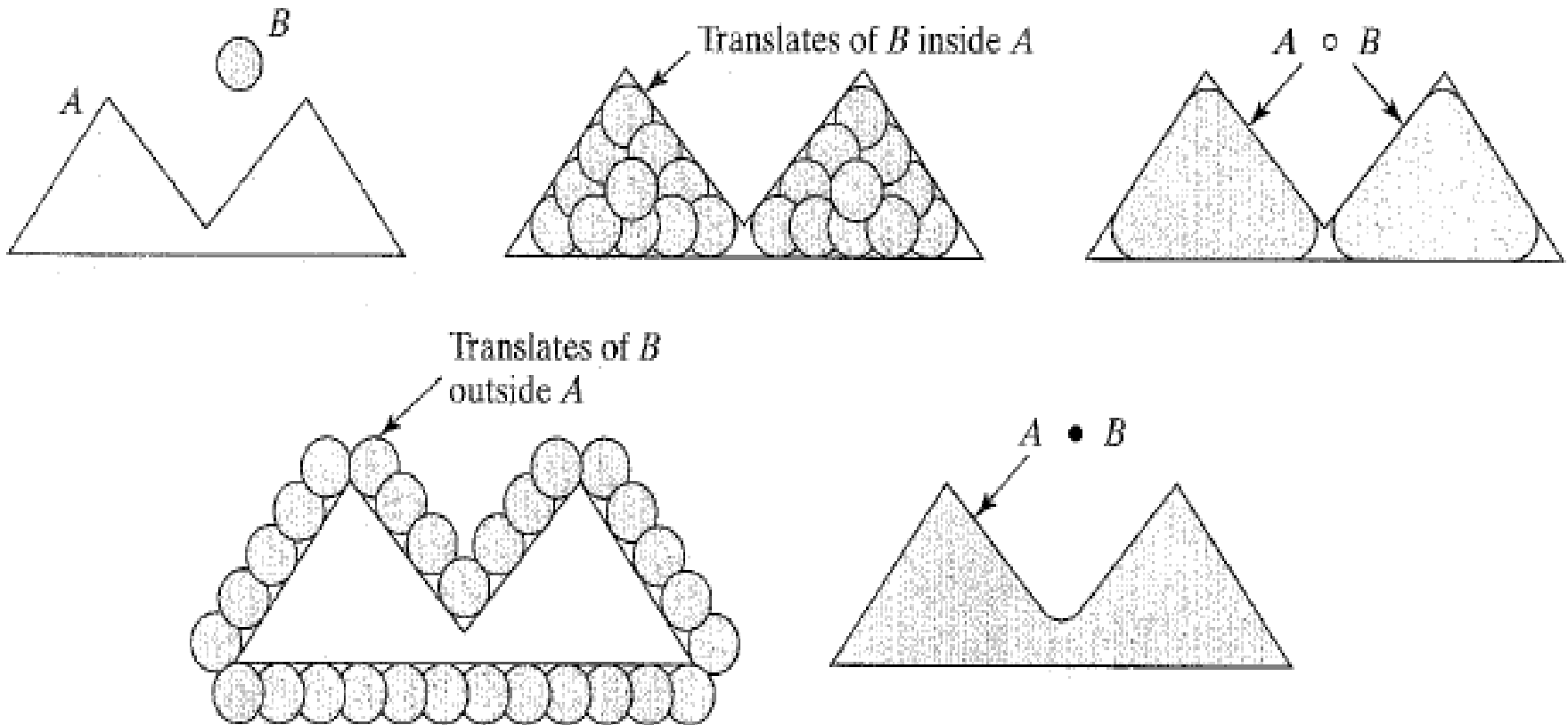
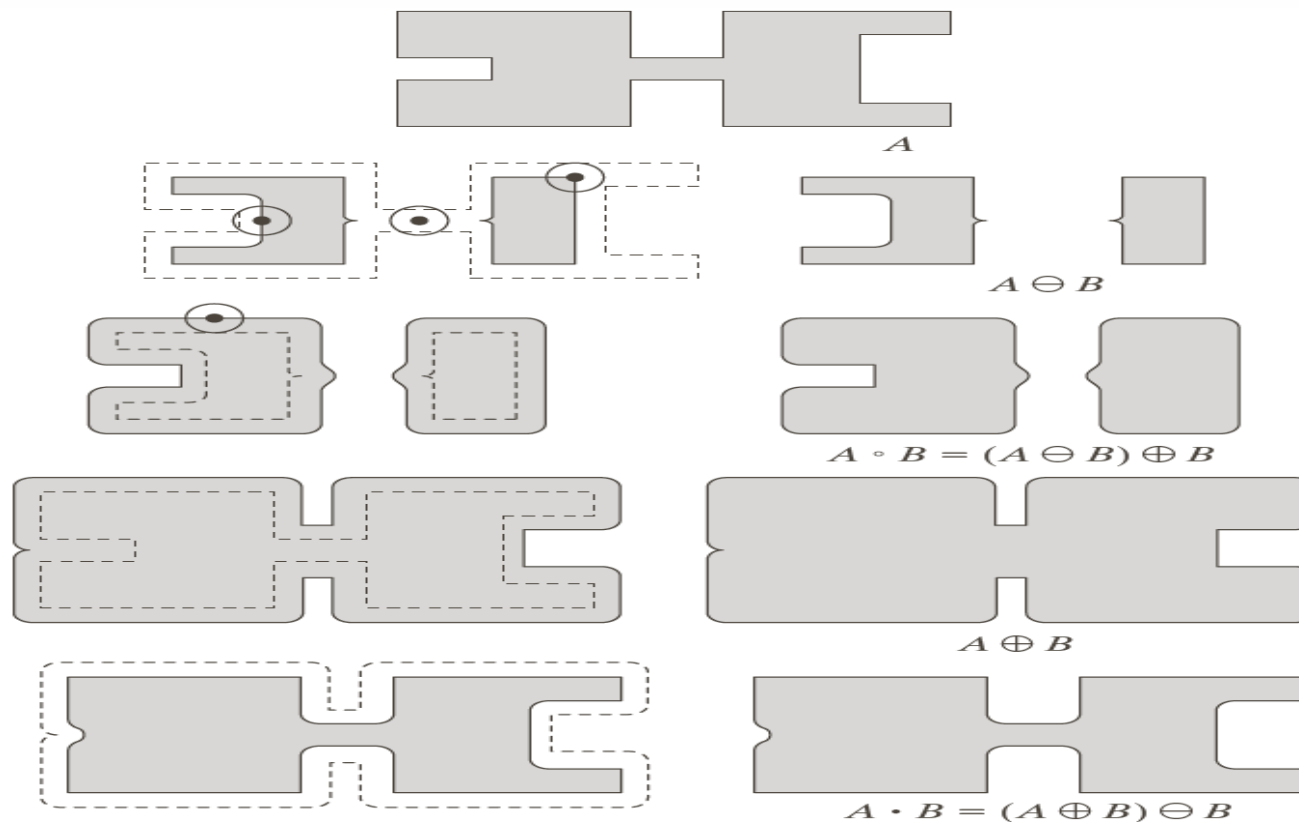


FIGURE 9.9 (a) Structuring element B “rolling” on the outer boundary of set A . (b) The heavy line is the outer boundary of the closing. (c) Complete closing (shaded). We did not shade A in (a) for clarity.





a	
b	c
d	e
f	g
h	i

FIGURE 9.10 Morphological opening and closing. The structuring element is the small circle shown in various positions in (b). The SE was not shaded here for clarity. The dark dot is the center of the structuring element.



A $A \ominus B$

$$B$$

1	1	1
1	1	1
1	1	1



$$(A \ominus B) \oplus B = A \circ B$$

$$(A \circ B) \oplus B$$

$$[(A \circ B) \oplus B] \ominus B = (A \circ B) \cdot B$$



Duality of Opening and Closing

- ❑ Opening and closing are duals of each other with respect to set complementation and reflection

PU

4.The Hit-or-Miss Transformation

The hit-or-miss transform is a general binary morphological operation that can be used to look for particular patterns of foreground and background pixels in an image.

The hit-and-miss transform is a basic tool for shape detection.

Concept: To detect a shape:

- ☐ **Hit object**
- ☐ **Miss background**

- ❑ Hit-or-Miss Transform
- ❑ The structural elements used for Hit-or-miss transforms are an extension to the ones used with dilation, erosion etc.
- ❑ The structural elements can contain both foreground and background pixels, rather than just foreground pixels, i.e. both ones and zeros.
- ❑ The structuring element is superimposed over each pixel in the input image, and if an exact match is found between the foreground and background pixels in the structuring element and the image, the input pixel lying below the origin of the structuring element is set to the foreground pixel value. If it does not match, the input pixel is replaced by the boundary pixel value.

The hit-or-miss transform is defined as:

Let $B = \{B_1, B_2\}$, where B_1 is the set formed from elements of B associated with an object and B_2 is the set of elements of B associated with the corresponding background, where B_1 and B_2 are disjoint

Example 1

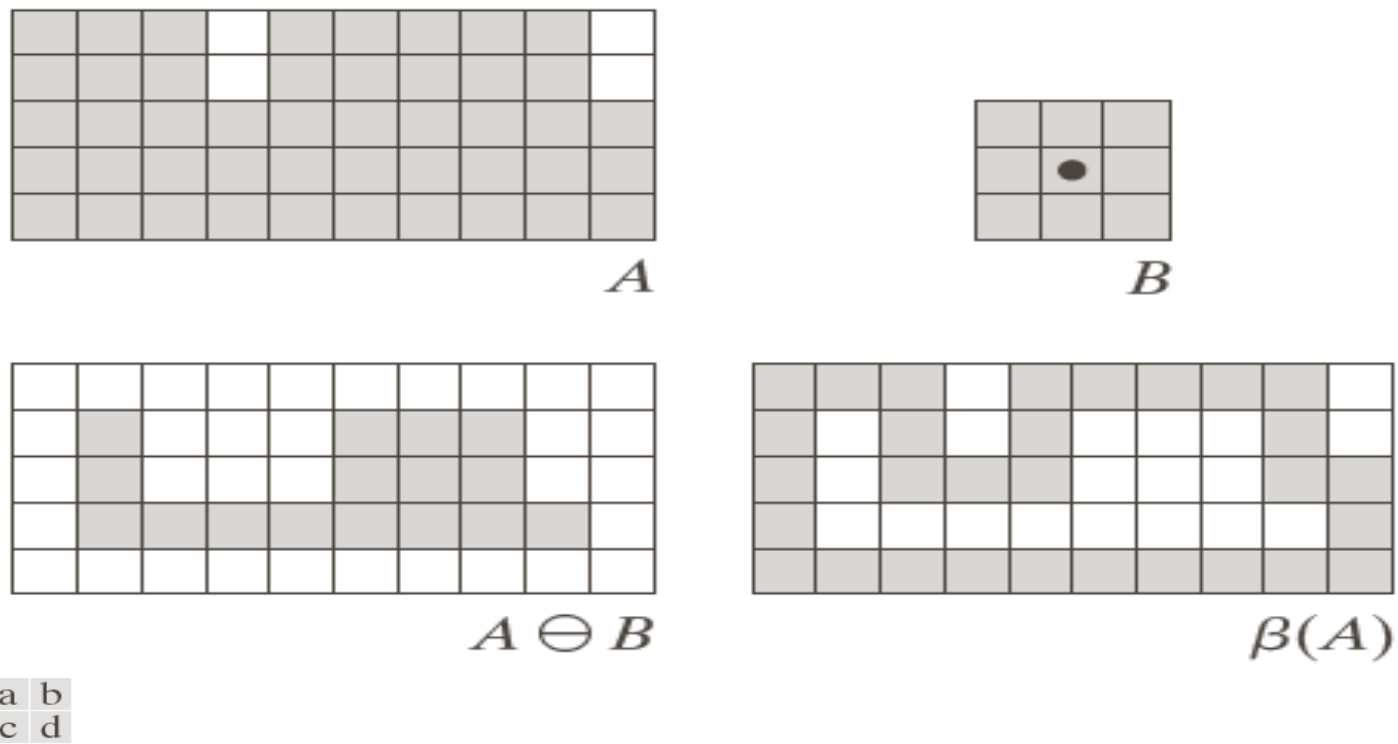


FIGURE 9.13 (a) Set A . (b) Structuring element B . (c) A eroded by B . (d) Boundary, given by the set difference between A and its erosion.

Example 2



a b

FIGURE 9.14

(a) A simple binary image, with 1s represented in white. (b) Result of using Eq. (9.5-1) with the structuring element in Fig. 9.13(b).

Some Basic Morphological Algorithms

❑ Hole Filling

A hole may be defined as a background region surrounded by a connected border of foreground pixels.

Let A denote a set whose elements are 8-connected boundaries, each boundary enclosing a background region (i.e., a hole). Given a point in each hole, the objective is to fill all the holes with 1s.

Some Basic Morphological Algorithms

❑ Hole Filling

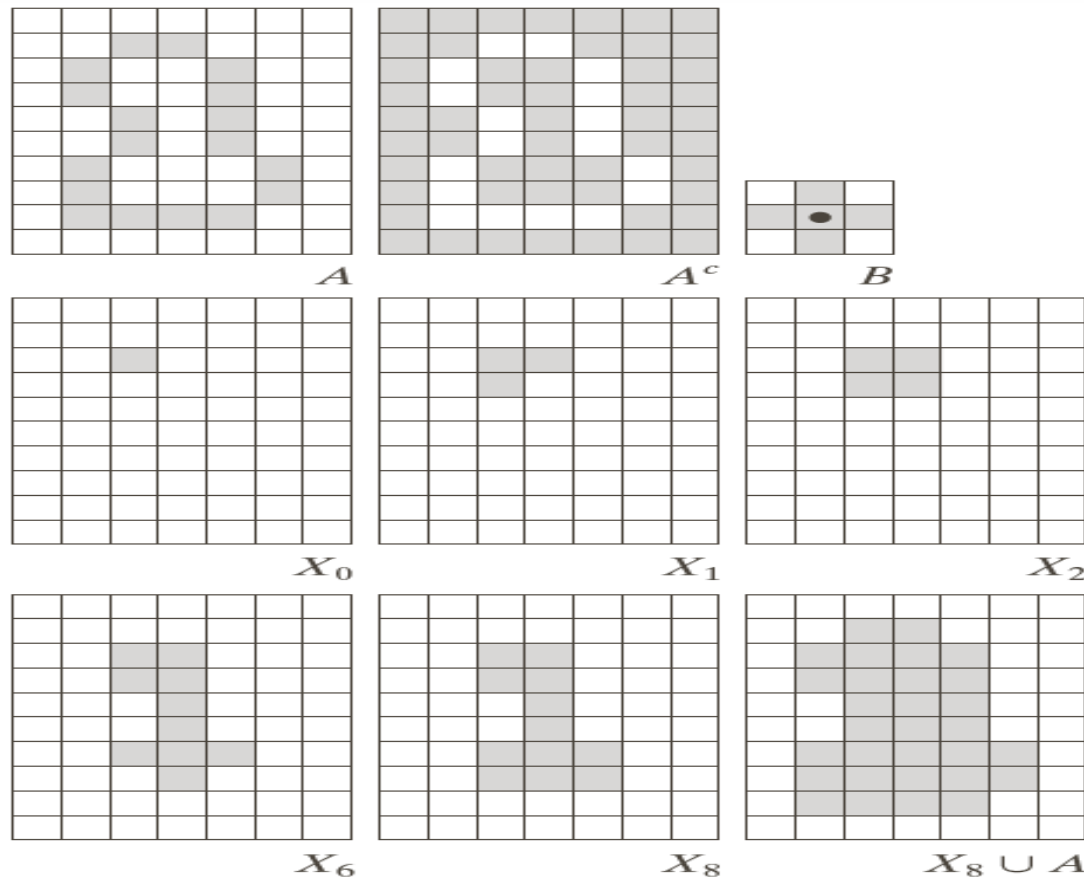
1. Forming an array X_0 of 0s (the same size as the array containing A), except the locations in X_0 corresponding to the given point in each hole, which we set to 1.

$$2. X_k = (X_{k-1} + B) \quad A^c \quad k=1,2,3,\dots$$

Stop the iteration if $X_k = X_{k-1}$



Example



a	b	c
d	e	f
g	h	i

FIGURE 9.15 Hole filling. (a) Set A (shown shaded). (b) Complement of A . (c) Structuring element B . (d) Initial point inside the boundary. (e)–(h) Various steps of Eq. (9.5-2). (i) Final result [union of (a) and (h)].

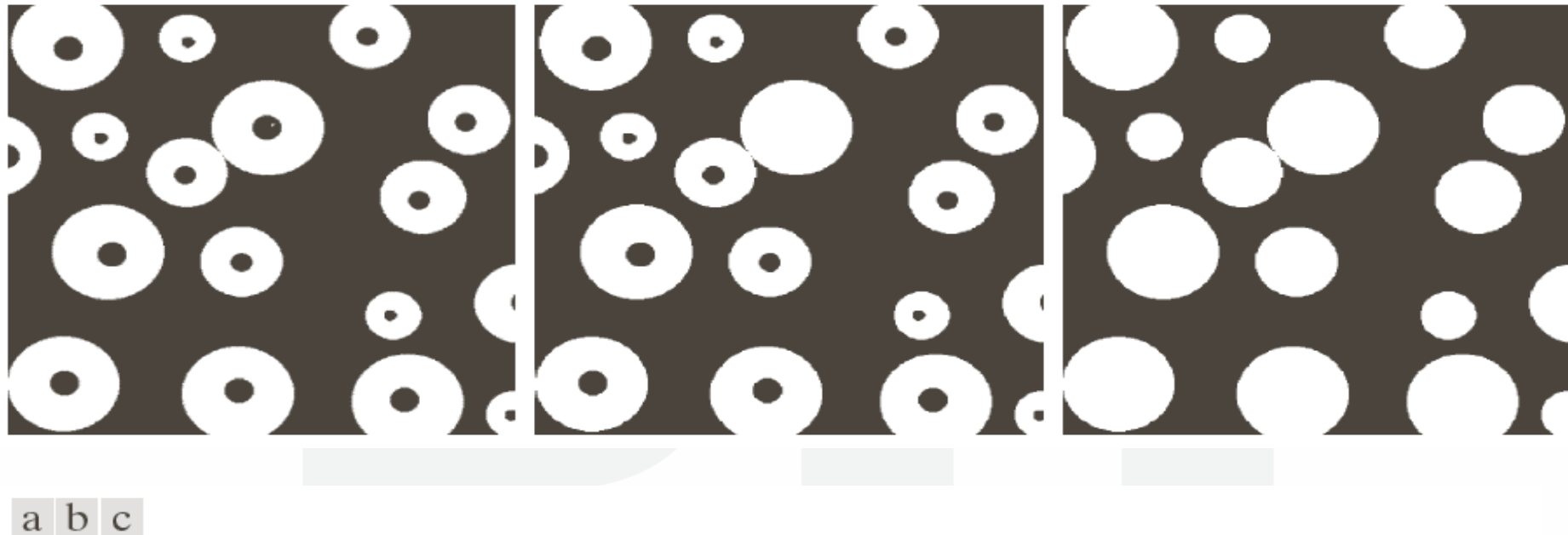


FIGURE 9.16 (a) Binary image (the white dot inside one of the regions is the starting point for the hole-filling algorithm). (b) Result of filling that region. (c) Result of filling all holes.

Some Basic Morphological Algorithms

❑ Extraction of Connected Components

Central to many automated image analysis applications.

Let A be a set containing one or more connected components, and form an array X_0 (of the same size as the array containing A) whose elements are 0s, except at each location known to correspond to a point in each connected component in A , which is set to 1.



Some Basic Morphological Algorithms

☐ **Extraction of Connected Components**

Central to many automated image analysis applications.

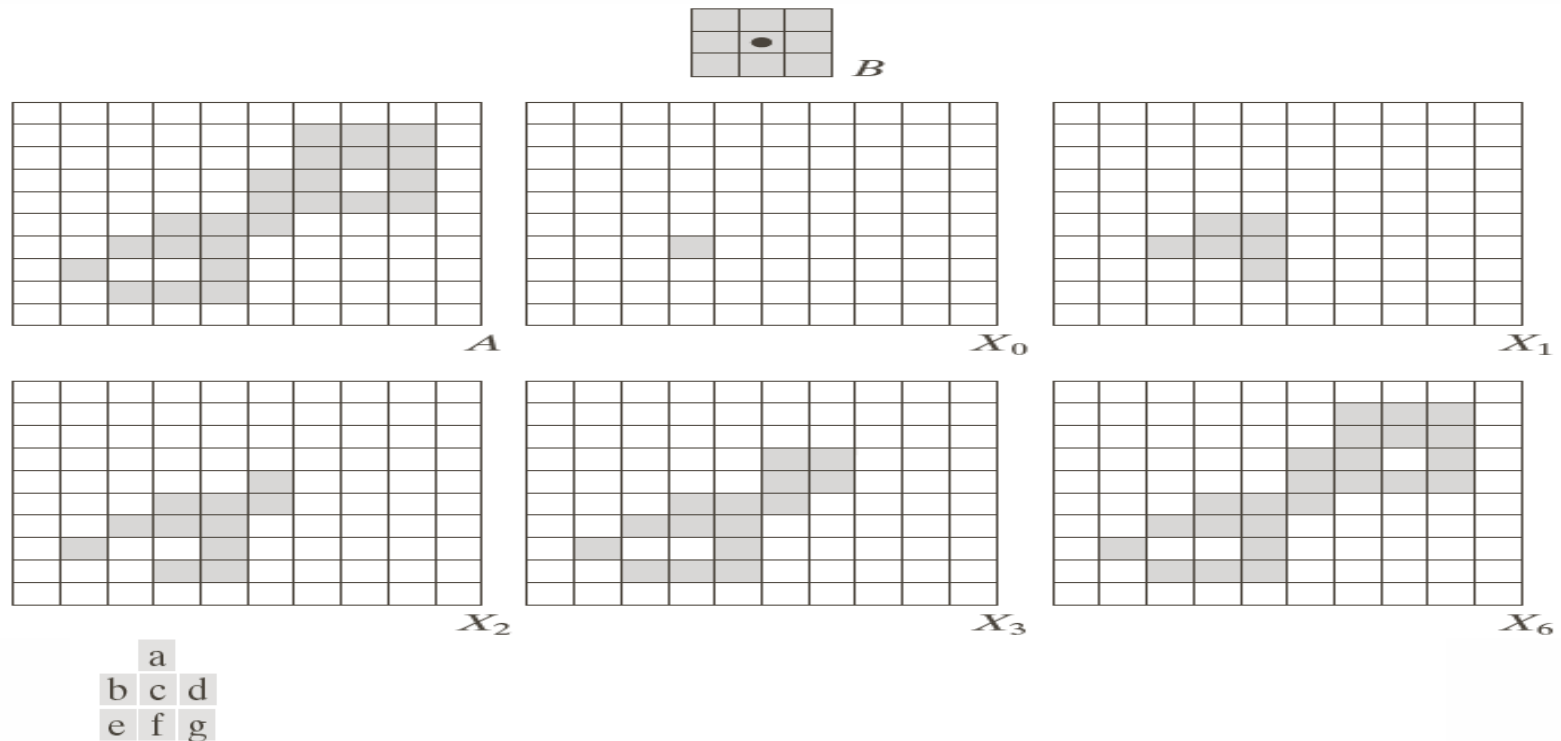
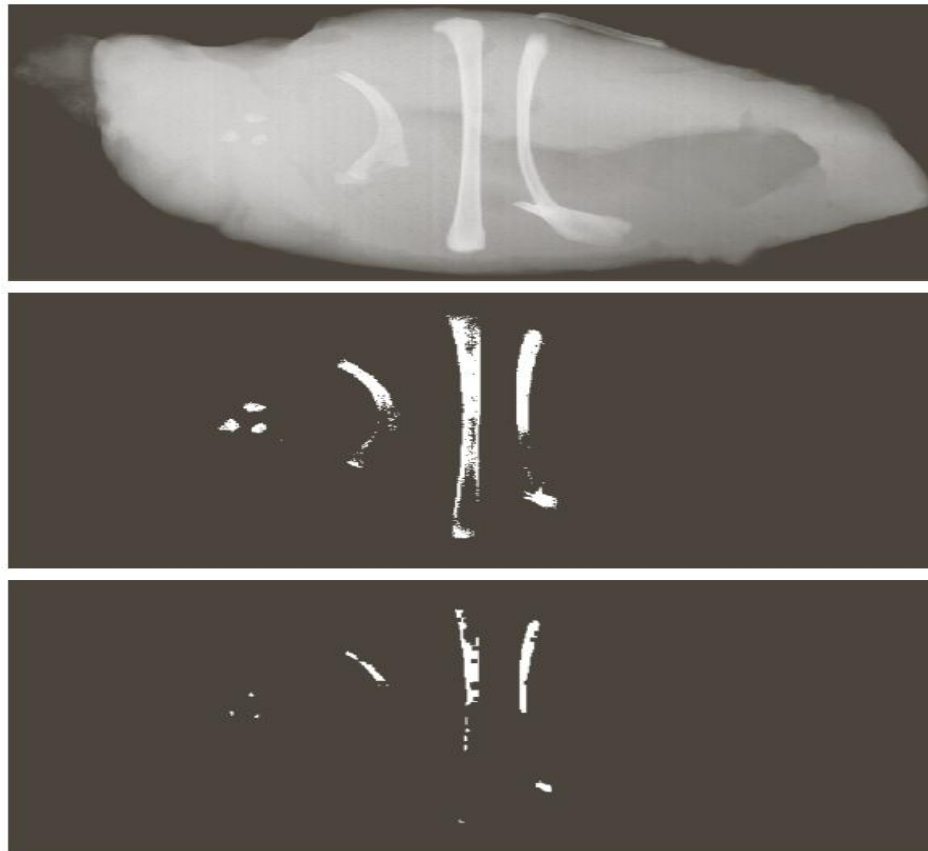


FIGURE 9.17 Extracting connected components. (a) Structuring element. (b) Array containing a set with one connected component. (c) Initial array containing a 1 in the region of the connected component. (d)–(g) Various steps in the iteration of Eq. (9.5-3).



a
b
c d

FIGURE 9.18
(a) X-ray image of chicken fillet with bone fragments.
(b) Thresholded image. (c) Image eroded with a 5×5 structuring element of 1s. (d) Number of pixels in the connected components of (c).
(Image courtesy of NTB Elektronische Geraete GmbH, Diepholz, Germany, www.ntbxray.com.)

Connected component	No. of pixels in connected comp
01	11
02	9
03	9
04	39
05	133
06	1
07	1
08	743
09	7
10	11
11	11
12	9
13	9
14	674
15	85

Some Basic Morphological Algorithms

□ Convex Hull

A set A is said to be **convex** if the straight line segment joining any two points in A lies entirely within A .

The **convex hull** H of an arbitrary set S is the smallest convex set containing S .

Some Basic Morphological Algorithms

❑ Convex Hull

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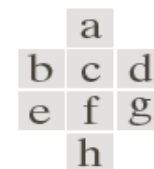
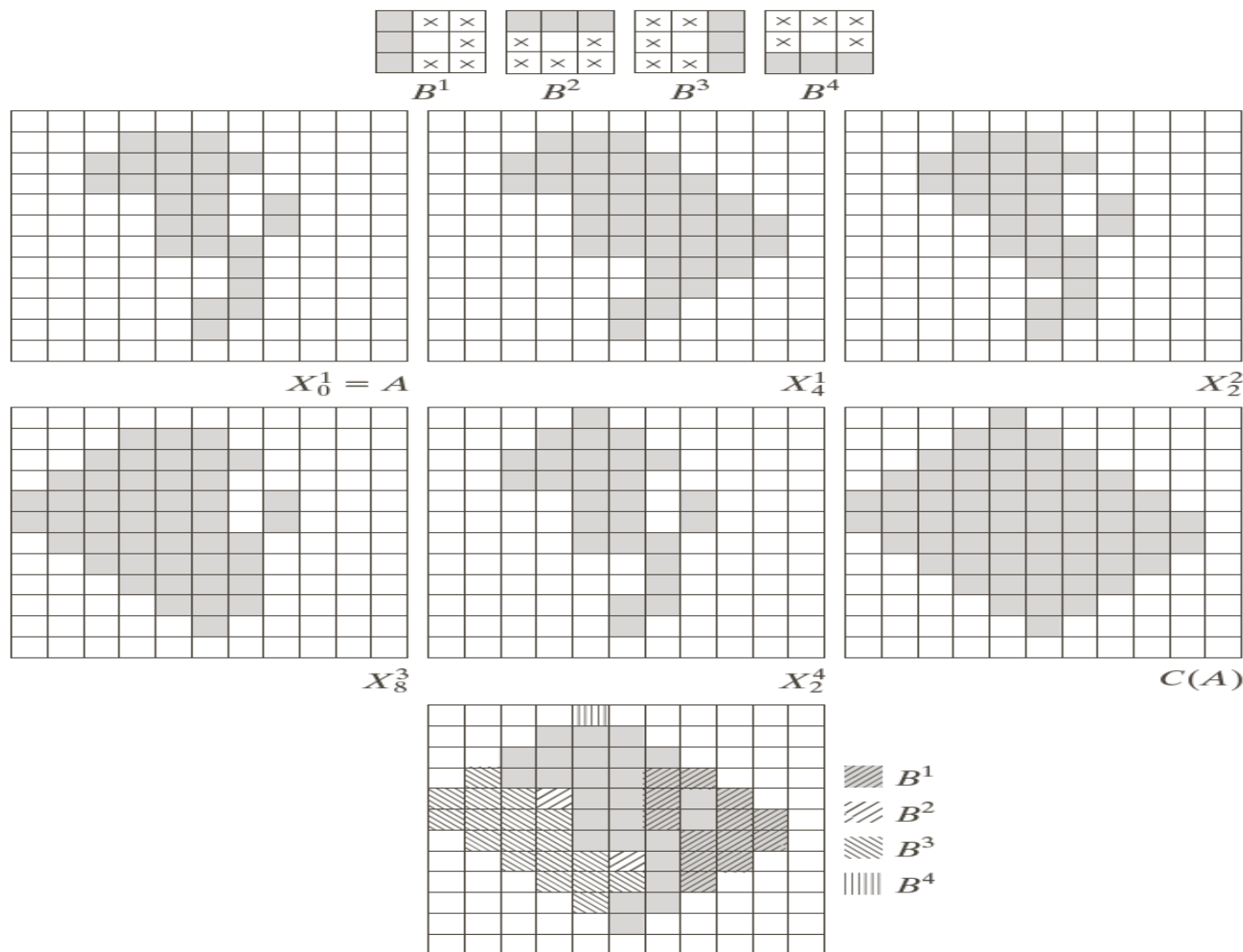


FIGURE 9.19
(a) Structuring elements. (b) Set A . (c)–(f) Results of convergence with the structuring elements shown in (a). (g) Convex hull. (h) Convex hull showing the contribution of each structuring element.

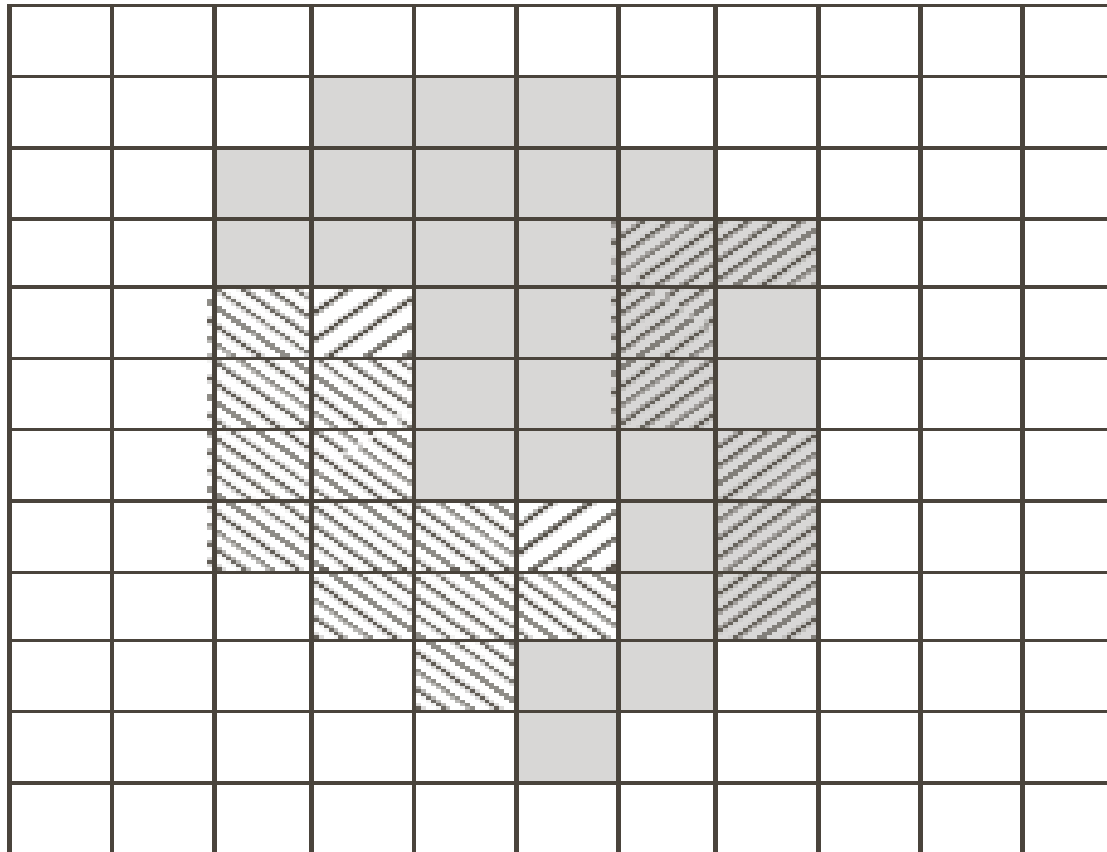


FIGURE 9.20

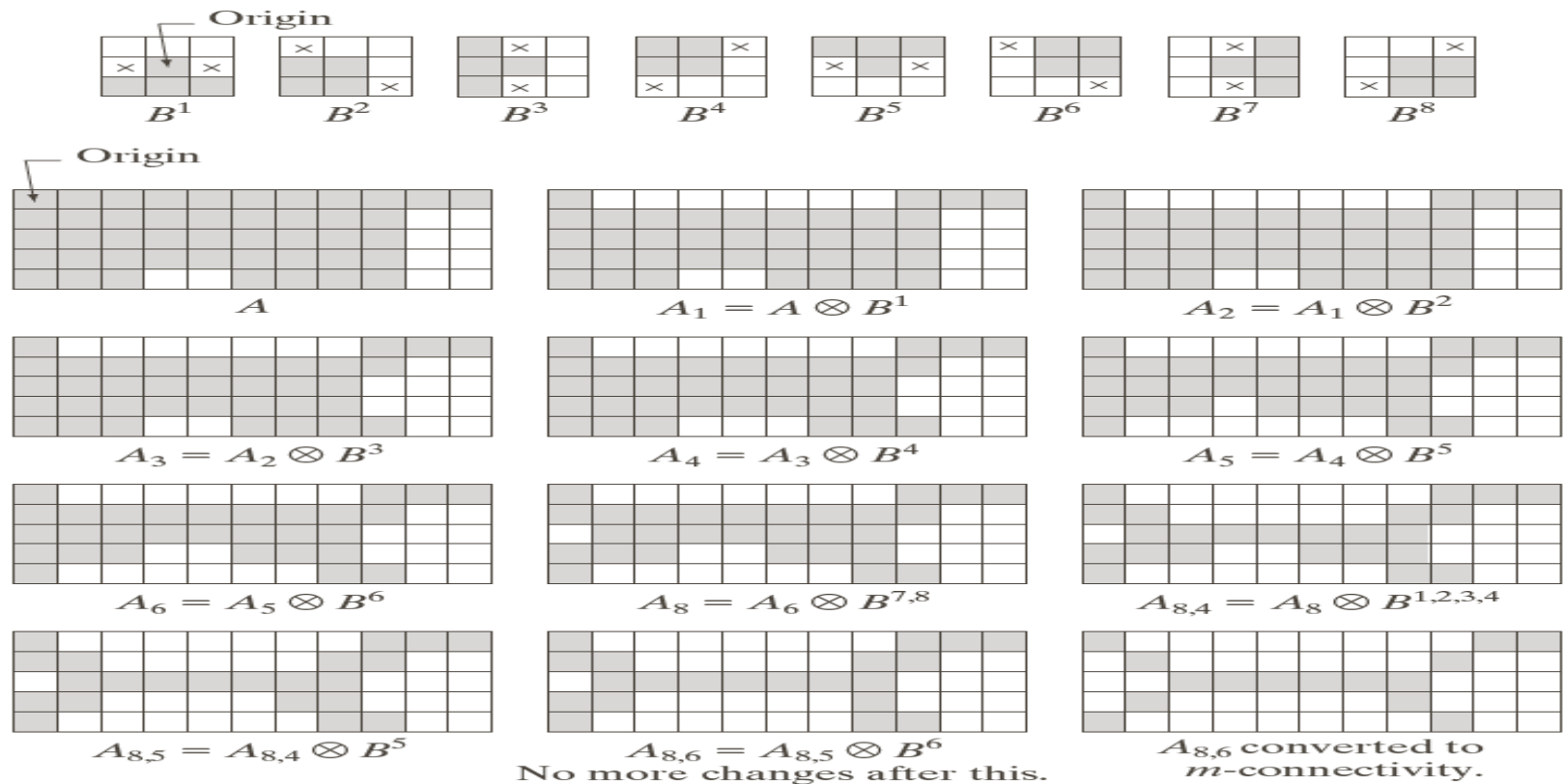
Result of limiting growth of the convex hull algorithm to the maximum dimensions of the original set of points along the vertical and horizontal directions.

Some Basic Morphological Algorithms

❑ Thinning

The thinning of a set A by a structuring element B , defined

PU



	a	
b	c	d
e	f	g
h	i	j
k	l	m

FIGURE 9.21 (a) Sequence of rotated structuring elements used for thinning. (b) Set A . (c) Result of thinning with the first element. (d)–(i) Results of thinning with the next seven elements (there was no change between the seventh and eighth elements). (j) Result of using the first four elements again. (l) Result after convergence. (m) Conversion to m -connectivity.

Some Basic Morphological Algorithms

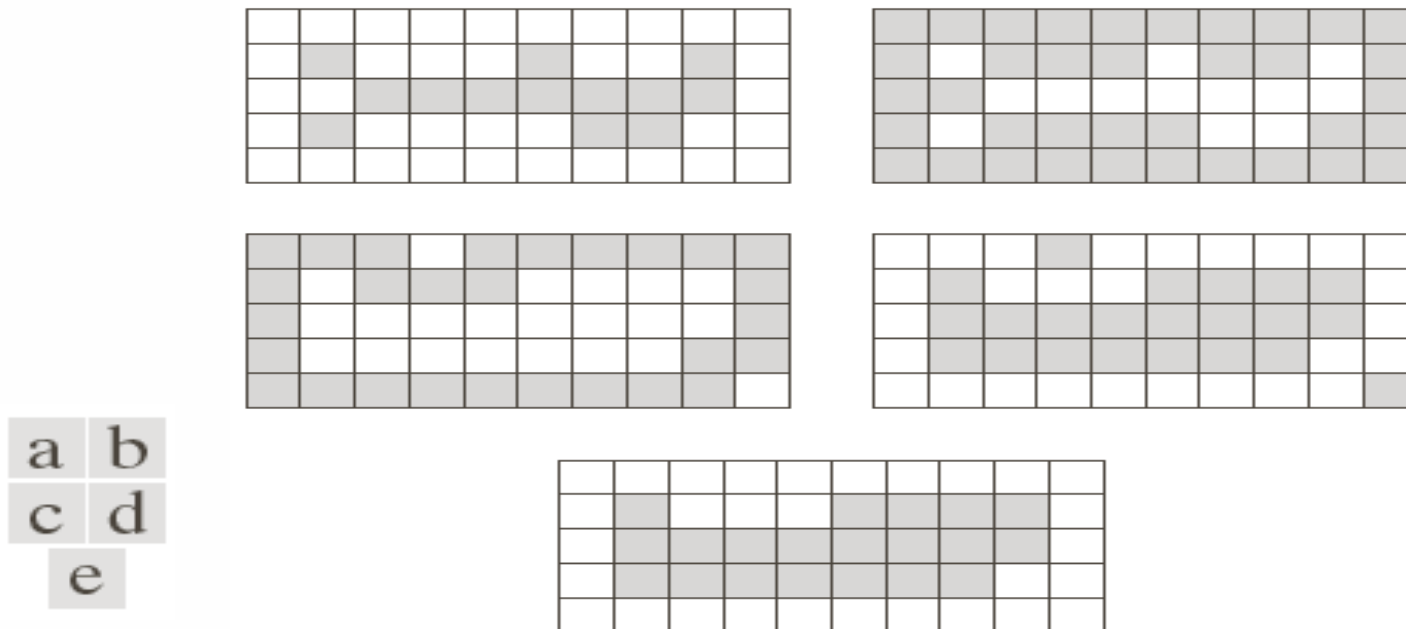


FIGURE 9.22 (a) Set A. (b) Complement of A. (c) Result of thinning the complement of A. (d) Thickened set obtained by complementing (c). (e) Final result, with no disconnected points.

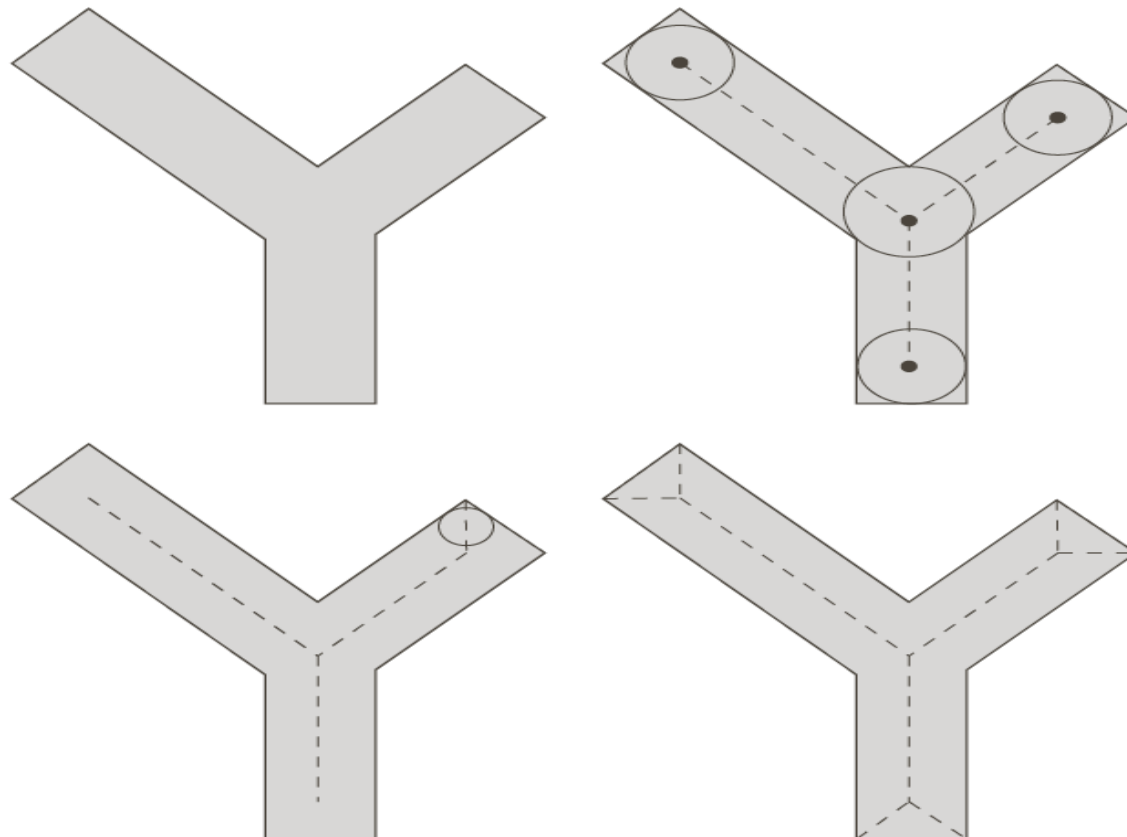


Some Basic Morphological Algorithms

❑ Skeletons

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Some Basic Morphological Algorithms



a	b
c	d

FIGURE 9.23

(a) Set A .
 (b) Various positions of maximum disks with centers on the skeleton of A .
 (c) Another maximum disk on a different segment of the skeleton of A .
 (d) Complete skeleton.

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k	$A \ominus kB$	$(A \ominus kB) \circ B$	$S_k(A)$	$\bigcup_{k=0}^K S_k(A)$	$S_k(A) \oplus kB$	$\bigcup_{k=0}^K S_k(A) \oplus kB$
0						
1						
2						

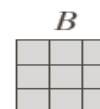


FIGURE 9.24

Implementation of Eqs. (9.5-11) through (9.5-15). The original set is at the top left, and its morphological skeleton is at the bottom of the fourth column. The reconstructed set is at the bottom of the sixth column.

Some Basic Morphological Algorithms

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k	$A \ominus kB$	$(A \ominus kB) \circ B$	$S_k(A)$	$\bigcup_{k=0}^K S_k(A)$	$S_k(A) \oplus kB$	$\bigcup_{k=0}^K S_k(A) \oplus kB$
0						
1						
2						

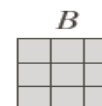


FIGURE 9.24

Implementation of Eqs. (9.5-11) through (9.5-15). The original set is at the top left, and its morphological skeleton is at the bottom of the fourth column. The reconstructed set is at the bottom of the sixth column.

Some Basic Morphological Algorithms

❑ Pruning

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Some Basic Morphological Algorithms

❖ Morphological Reconstruction

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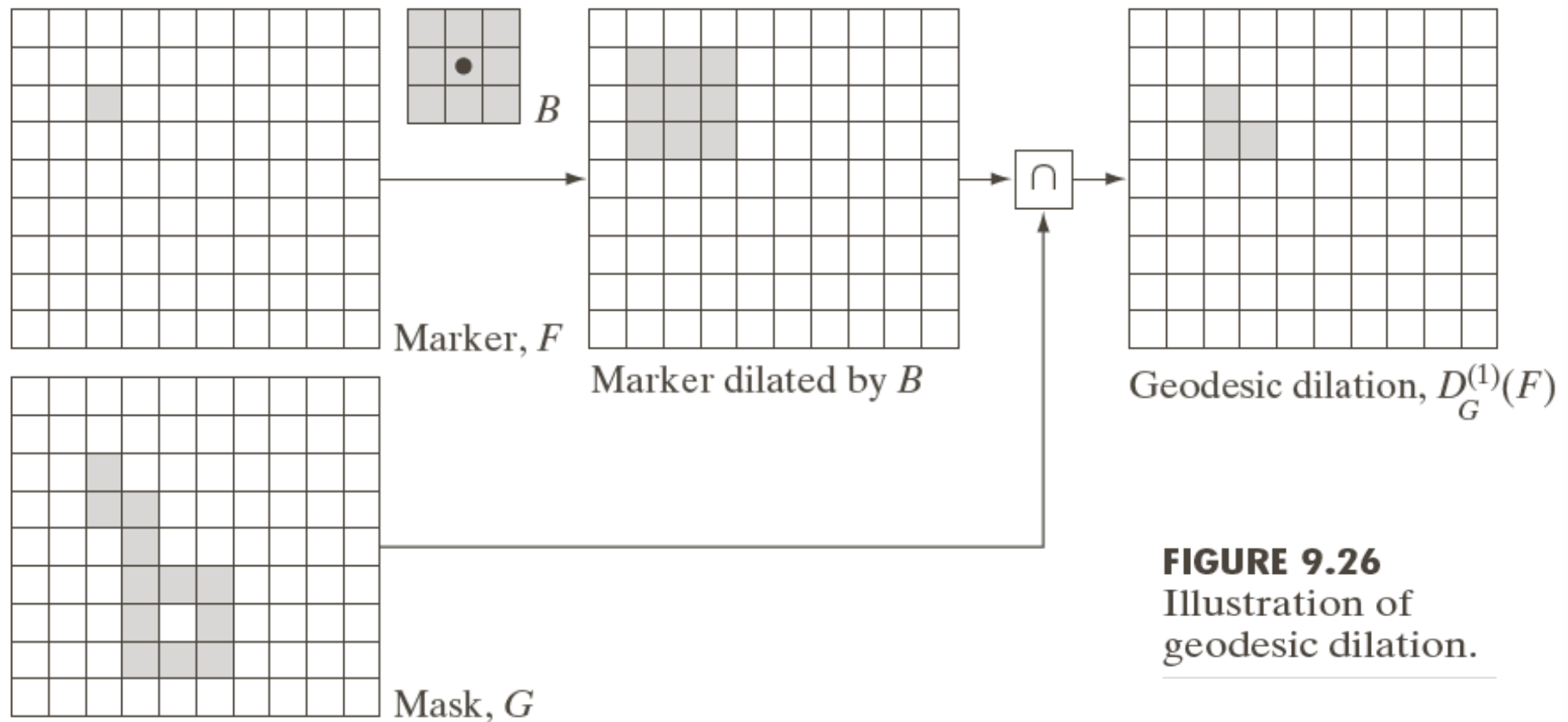


FIGURE 9.26
Illustration of
geodesic dilation.

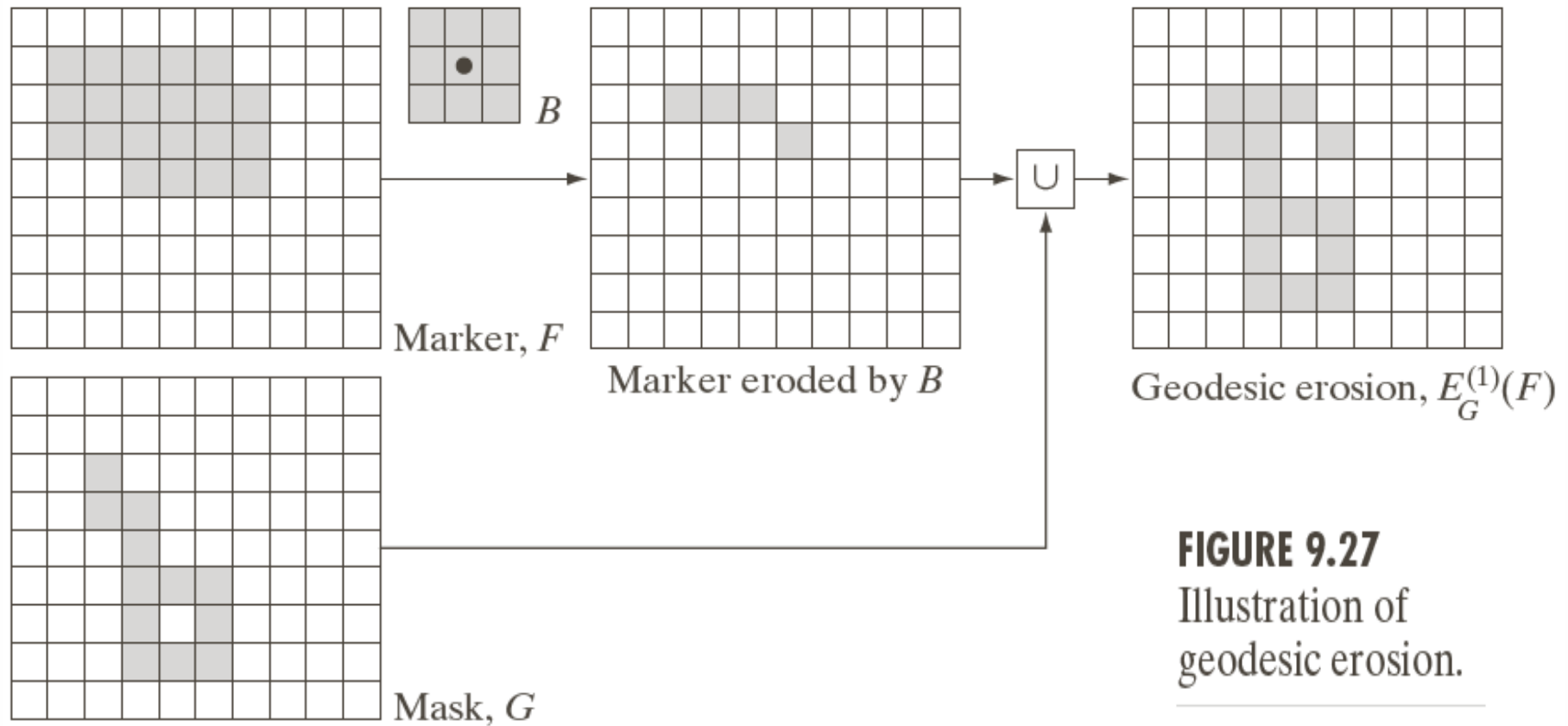
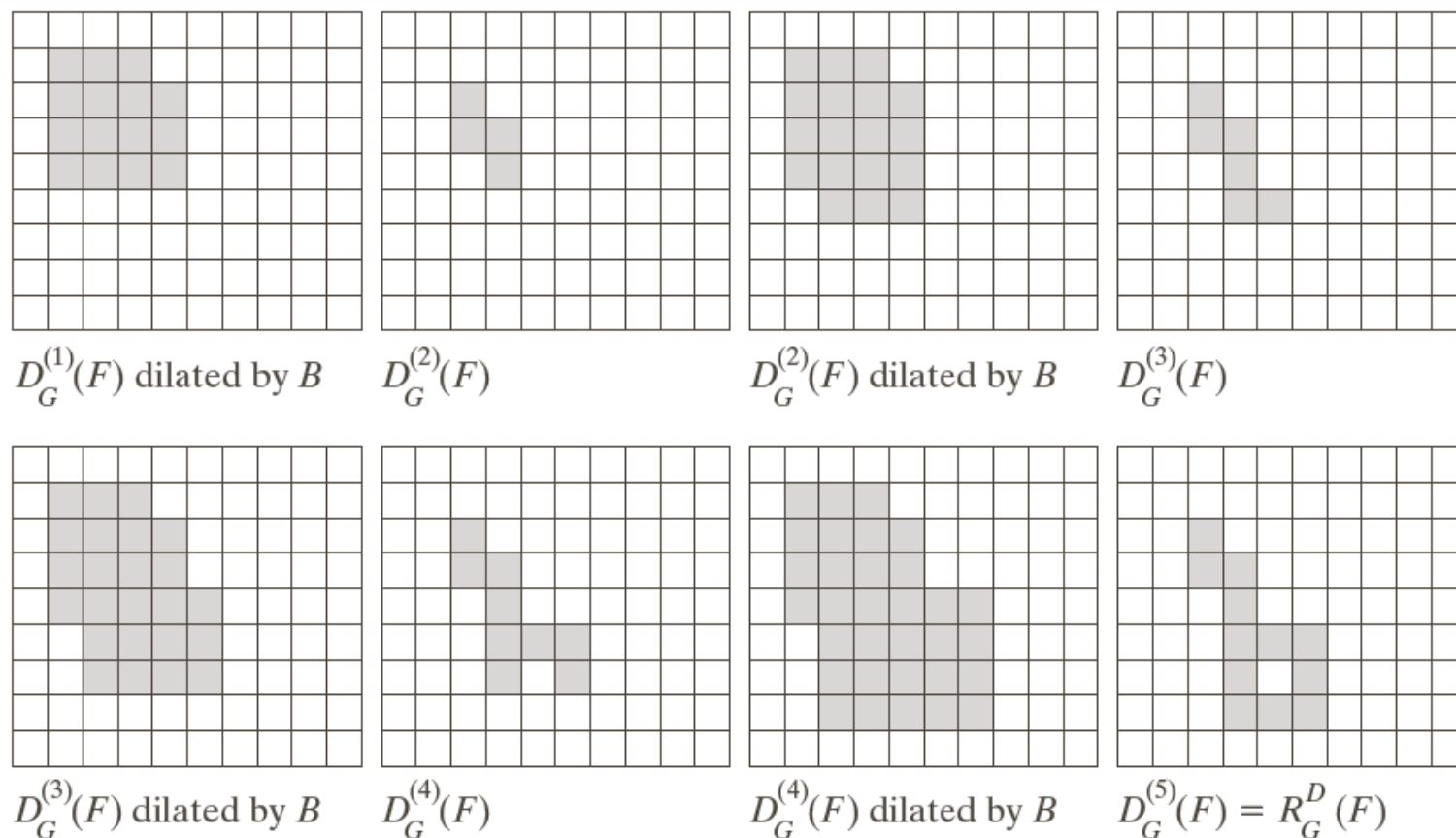


FIGURE 9.27
Illustration of
geodesic erosion.

Morphological Reconstruction by Dilation

PU



a	b	c	d
e	f	g	h

FIGURE 9.28
Illustration of morphological reconstruction by dilation. F , G , B and $D_G^{(1)}(F)$ are from Fig. 9.26.

Morphological Reconstruction by Erosion

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Sample Applications

Morphological reconstruction has broad spectrum of practical applications: Each determined by the selection of the marker and mask images, by the structuring elements used by the combination of primitive operations discussed in preceding sections:

- Opening by Reconstruction
- Filling Holes
- Border Clearing

Opening by Reconstruction

In a morphological opening, erosion removes small objects and the subsequent dilation attempts to restore the shape of the objects that remain. However, the accuracy of this restoration is highly dependent on the similarity of the shapes of the objects and the structuring element used.

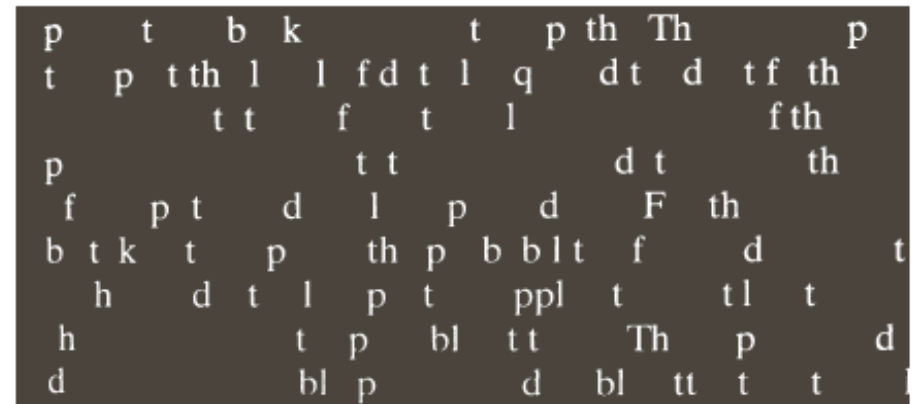
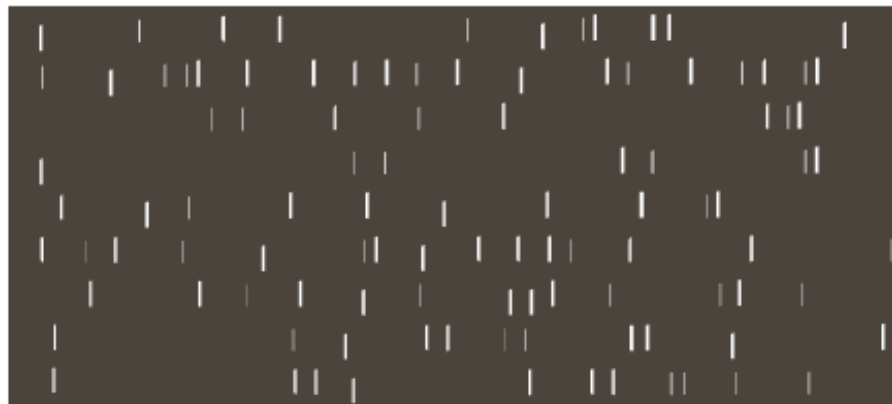
Opening by reconstruction restores exactly the shapes of the objects that remain after erosion

Opening by Reconstruction

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ponents or broken connection paths. There is no point in going past the level of detail required to identify those components.

Segmentation of nontrivial images is one of the most difficult tasks in image processing. Segmentation accuracy determines the effectiveness of computerized analysis procedures. For this reason, considerable effort can be taken to improve the probability of rugged segmentation. In applications such as industrial inspection applications, at least some degree of segmentation of the environment is possible at times. The experienced image processing designer invariably pays considerable attention to such

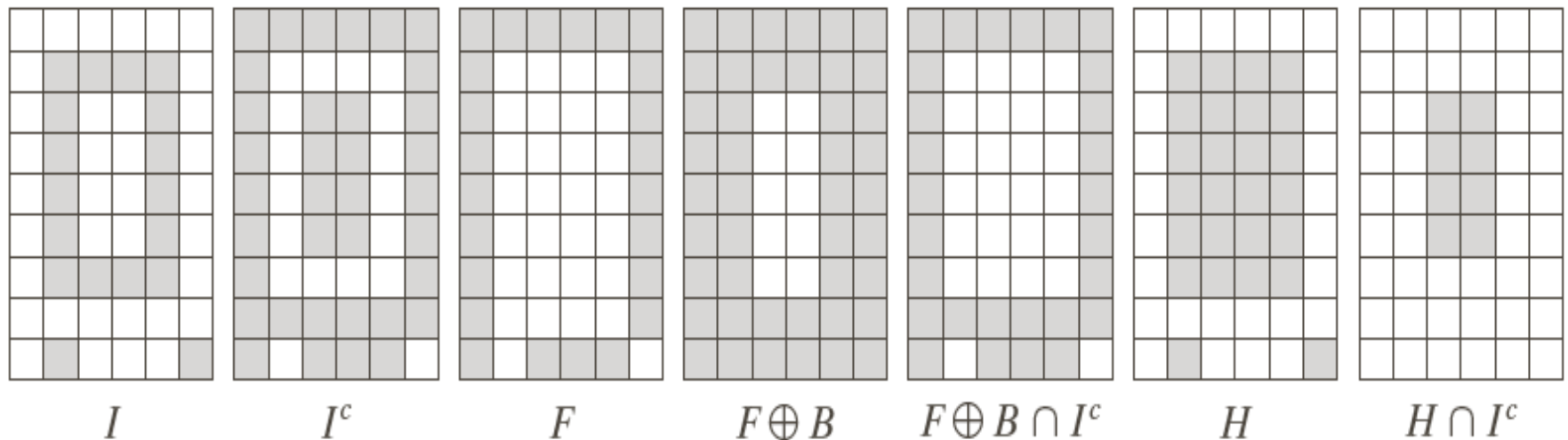


a	b
c	d

FIGURE 9.29 (a) Text image of size 918×2018 pixels. The approximate average height of the tall characters is 50 pixels. (b) Erosion of (a) with a structuring element of size 51×1 pixels. (c) Opening of (a) with the same structuring element, shown for reference. (d) Result of opening by reconstruction.

Filling Holes

PU



a b c d e f g

FIGURE 9.30
Illustration of
hole filling on a
simple image.



ponents or broken connection paths. There is no point in going past the level of detail required to identify those components.

Segmentation of nontrivial images is one of the most difficult tasks in image processing. Segmentation accuracy determines the effectiveness of computerized analysis procedures. For this reason, care must be taken to improve the probability of rugged segmentation. In applications such as industrial inspection applications, at least some level of segmentation in the environment is possible at times. The experienced image designer invariably pays considerable attention to such

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Segmentation of nontrivial images is one of the most difficult tasks in image processing. Segmentation accuracy determines the effectiveness of computerized analysis procedures. For this reason, care must be taken to improve the probability of rugged segmentation. In applications such as industrial inspection applications, at least some level of segmentation in the environment is possible at times. The experienced image designer invariably pays considerable attention to such

a	b
c	d

FIGURE 9.31

(a) Text image of size 918×2018 pixels. (b) Complement of (a) for use as a mask image. (c) Marker image. (d) Result of hole-filling using Eq. (9.5-29).

Border Clearing

PU



ponents or broken connection paths. There is no position past the level of detail required to identify those

Segmentation of nontrivial images is one of the most difficult tasks in image processing. Segmentation accuracy determines the effectiveness of computerized analysis procedures. For this reason, considerable effort can be taken to improve the probability of rugged segmentation, such as industrial inspection applications, at least some of the time. The experienced designer invariably pays considerable attention to such

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Segmentation of nontrivial images is one of the most difficult tasks in image processing. Segmentation accuracy determines the effectiveness of computerized analysis procedures. For this reason, considerable effort can be taken to improve the probability of rugged segmentation, such as industrial inspection applications, at least some of the time. The experienced designer invariably pays considerable attention to such

a b

FIGURE 9.32

Border clearing.
(a) Marker image.
(b) Image with no objects touching the border. The original image is Fig. 9.29(a).

Summary

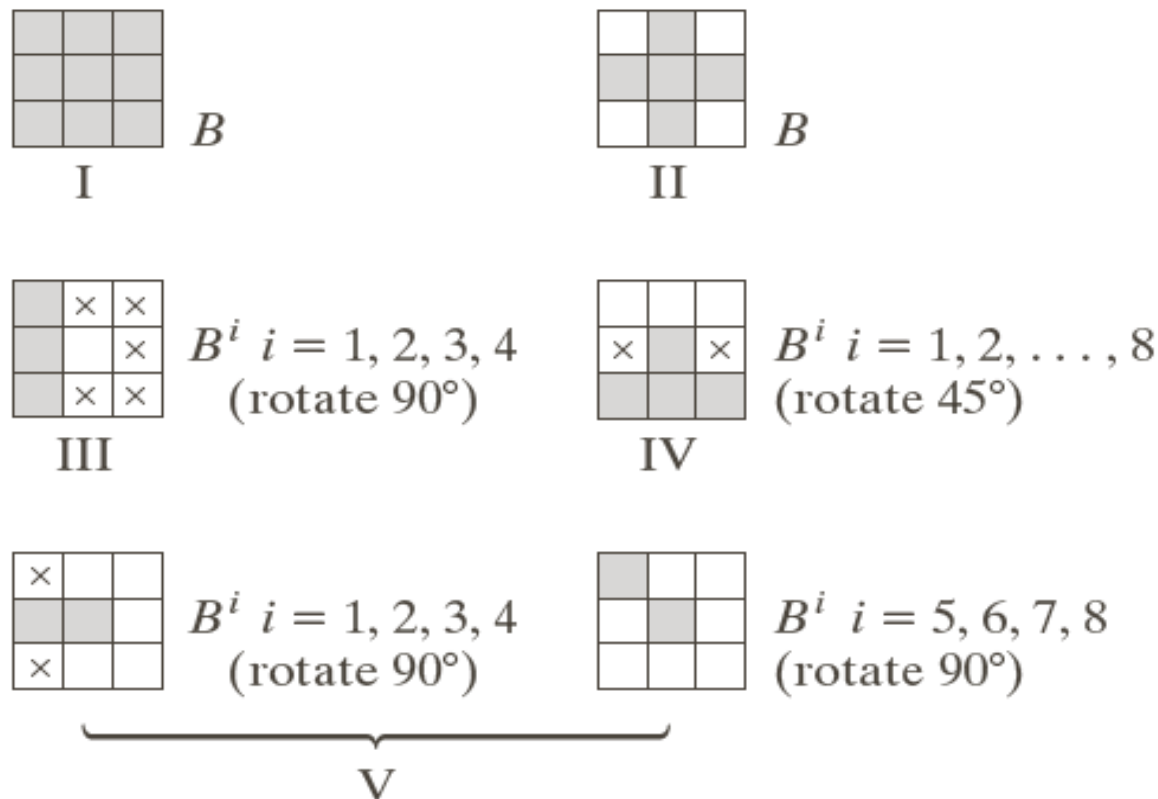


FIGURE 9.33 Five basic types of structuring elements used for binary morphology. The origin of each element is at its center and the X's indicate "don't care" values.

Summary

		Comments
Operation	Equation	(The Roman numerals refer to the structuring elements in Fig. 9.33.)
Translation	$(B)_z = \{w w = b + z, \text{ for } b \in B\}$	Translates the origin of B to point z .
Reflection	$\hat{B} = \{w w = -b, \text{ for } b \in B\}$	Reflects all elements of B about the origin of this set.
Complement	$A^c = \{w w \notin A\}$	Set of points not in A .
Difference	$A - B = \{w w \in A, w \notin B\}$ $= A \cap B^c$	Set of points that belong to A but not to B .
Dilation	$A \oplus B = \{z (\hat{B}_z) \cap A \neq \emptyset\}$	“Expands” the boundary of A . (I)
Erosion	$A \ominus B = \{z (B)_z \subseteq A\}$	“Contracts” the boundary of A . (I)
Opening	$A \circ B = (A \ominus B) \oplus B$	Smooths contours, breaks narrow isthmuses, and eliminates small islands and sharp peaks. (I)

TABLE 9.1
Summary of morphological operations and their properties.

(Continued)



TABLE 9.1
(Continued)

Operation	Equation	Comments (The Roman numerals refer to the structuring elements in Fig. 9.33.)
Closing	$A \bullet B = (A \oplus B) \ominus B$	Smooths contours, fuses narrow breaks and long thin gulfs, and eliminates small holes. (I)
Hit-or-miss transform	$A \otimes B = (A \ominus B_1) \cap (A^c \ominus B_2)$ $= (A \ominus B_1) - (A \oplus \tilde{B}_2)$	The set of points (coordinates) at which, simultaneously, B_1 found a match ("hit") in A and B_2 found a match in A^c
Boundary extraction	$\beta(A) = A - (A \ominus B)$	Set of points on the boundary of set A . (I)
Hole filling	$X_k = (X_{k-1} \oplus B) \cap A^c$; $k = 1, 2, 3, \dots$	Fills holes in A ; X_0 = array of 0s with a 1 in each hole. (II)
Connected components	$X_k = (X_{k-1} \oplus B) \cap A$; $k = 1, 2, 3, \dots$	Finds connected components in A ; X_0 = array of 0s with a 1 in each connected component. (I)
Convex hull	$X_k^i = (X_{k-1}^i \otimes B^i) \cup A$; $i = 1, 2, 3, 4$; $k = 1, 2, 3, \dots$; $X_0^i = A$; and $D^i = X_{\text{conv}}^i$	Finds the convex hull $C(A)$ of set A , where "conv" indicates convergence in the sense that $X_k^i = X_{k-1}^i$. (III)
Thinning	$A \otimes B = A - (A \otimes B)$ $= A \cap (A \otimes B)^c$ $A \otimes \{B\} =$ $((\dots ((A \otimes B^1) \otimes B^2) \dots) \otimes B^n)$ $\{B\} = \{B^1, B^2, B^3, \dots, B^n\}$	Thins set A . The first two equations give the basic definition of thinning. The last equations denote thinning by a sequence of structuring elements. This method is normally used in practice. (IV)
Thickening	$A \odot B = A \cup (A \otimes B)$ $A \odot \{B\} =$ $((\dots (A \odot B^1) \odot B^2 \dots) \odot B^n)$	Thickens set A . (See preceding comments on sequences of structuring elements.) Uses IV with 0s and 1s reversed.
Skeletons	$S(A) = \bigcup_{k=0}^K S_k(A)$ $S_k(A) = \bigcup_{k=0}^K \{(A \ominus kB) - [(A \ominus kB) \odot B]\}$ Reconstruction of A : $A = \bigcup_{k=0}^K (S_k(A) \oplus kB)$	Finds the skeleton $S(A)$ of set A . The last equation indicates that A can be reconstructed from its skeleton subsets $S_k(A)$. In all three equations, K is the value of the iterative step after which the set A erodes to the empty set. The notation $(A \ominus kB)$ denotes the k th iteration of successive erosions of A by B . (I)

(Continued)



Operation		Equation	Comments
Pruning		$X_1 = A \otimes \{B\}$ $X_2 = \bigcup_{k=1}^8 (X_1 \otimes B^k)$ $X_3 = (X_2 \oplus H) \cap A$ $X_4 = X_1 \cup X_3$	(The Roman numerals refer to the structuring elements in Fig. 9.33.) X_4 is the result of pruning set A. The number of times that the first equation is applied to obtain X_1 must be specified. Structuring elements V are used for the first two equations. In the third equation H denotes structuring element I. F and G are called the <i>marker</i> and <i>mask</i> images, respectively.
Geodesic dilation of size 1		$D_G^{(1)}(F) = (F \oplus B) \cap G$	
Geodesic dilation of size n		$D_G^{(n)}(F) = D_G^{(1)}[D_G^{(n-1)}(F)];$ $D_G^{(0)}(F) = F$	
Geodesic erosion of size 1		$E_G^{(1)}(F) = (F \ominus B) \cup G$	
Geodesic erosion of size n		$E_G^{(n)}(F) = E_G^{(1)}[E_G^{(n-1)}(F)];$ $E_G^{(0)}(F) = F$	
Morphological reconstruction by dilation		$R_G^D(F) = D_G^{(k)}(F)$	k is such that $D_G^{(k)}(F) = D_G^{(k+1)}(F)$
Morphological reconstruction by erosion		$R_G^E(F) = E_G^{(k)}(F)$	k is such that $E_G^{(k)}(F) = E_G^{(k+1)}(F)$
Opening by reconstruction		$O_R^{(n)}(F) = R_F^D[(F \ominus nB)]$	$(F \ominus nB)$ indicates n erosions of F by B .
Closing by reconstruction		$C_R^{(n)}(F) = R_F^E[(F \oplus nB)]$	$(F \oplus nB)$ indicates n dilations of F by B .
Hole filling		$H = [R_F^D(F)]^c$	H is equal to the input image I , but with all holes filled. See Eq. (9.5-28) for the definition of the marker image F .
Border clearing		$X = I - R_I^D(F)$	X is equal to the input image I , but with all objects that touch (are connected to) the boundary removed. See Eq. (9.5-30) for the definition of the marker image F .

TABLE 9.1
(Continued)

References

- ❑ Digital Image Processing by Rafael C. Gonzalez and Richard E. Woods, Pearson Education Publisher.
- ❑ Various Internet Resources

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